

# The Role of AI

Algorithm to bedside symposia





# DATA STORIES

## From algorithm to bedside

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Senior Forsker

Gentofte or Herlev Hospital

03-14 Feb. 2025

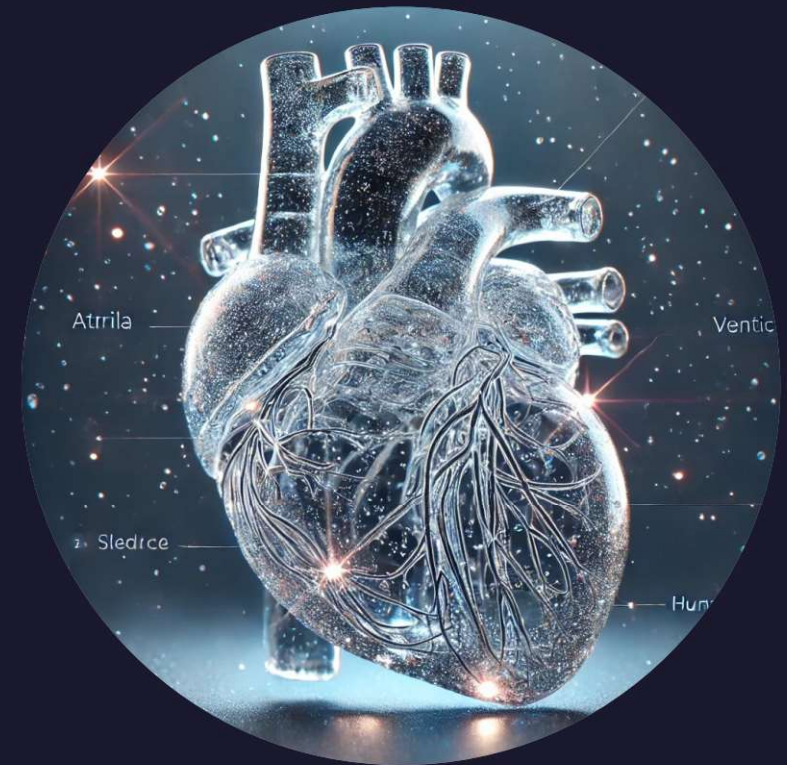
# Lunch symposia

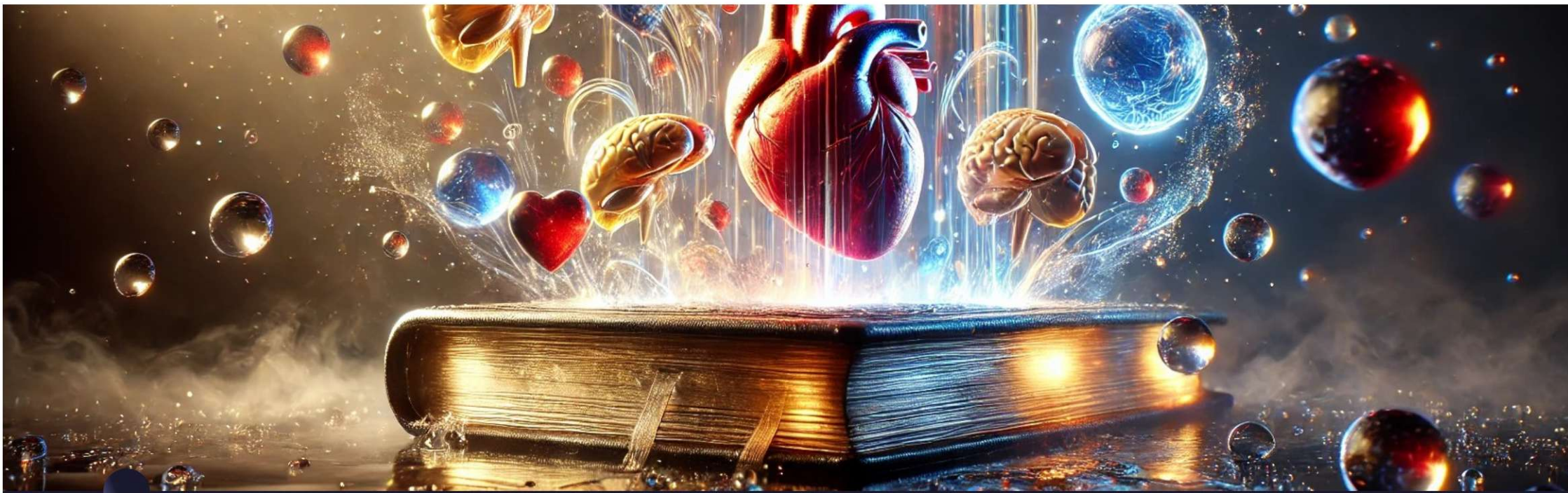
Omics-Based Clinical and Population Studies (session 1:  
Mon. 3 Feb. 2025 - Specific Case Studies)

Contextual Epidemiology of Cardiometabolic  
Conditions, Polypharmacy and Multi-morbidity  
(session 2: Thurs. 6 Feb. 2025 )

Scientific Methods in Biomedical Research (sessions 3  
and 4: Mon and Thurs. 10 and 13 Feb. 2025)

- Omics in biomedical research
- Evidence and analytics
- Global Burden of Disease, Health Analytics, and Access to Therapeutics
- Economic Evaluations





# Session 3

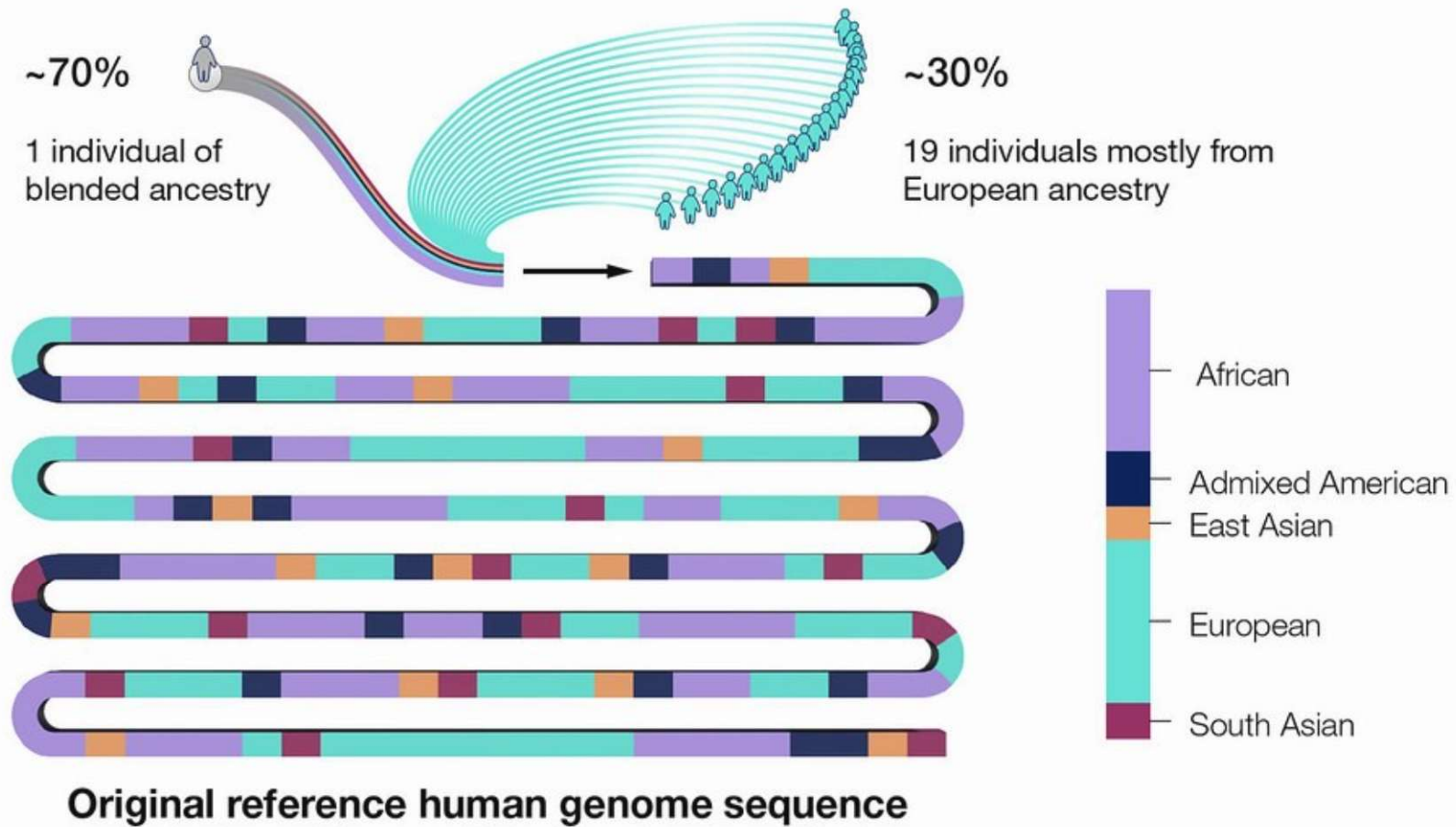
- Omics in biomedical research – **AI-based dry labs.**
- Evidence pyramid and analytics – **clinical vs. AI/omics-based studies**

# From algorithm to bedside

Omics  
pipelines

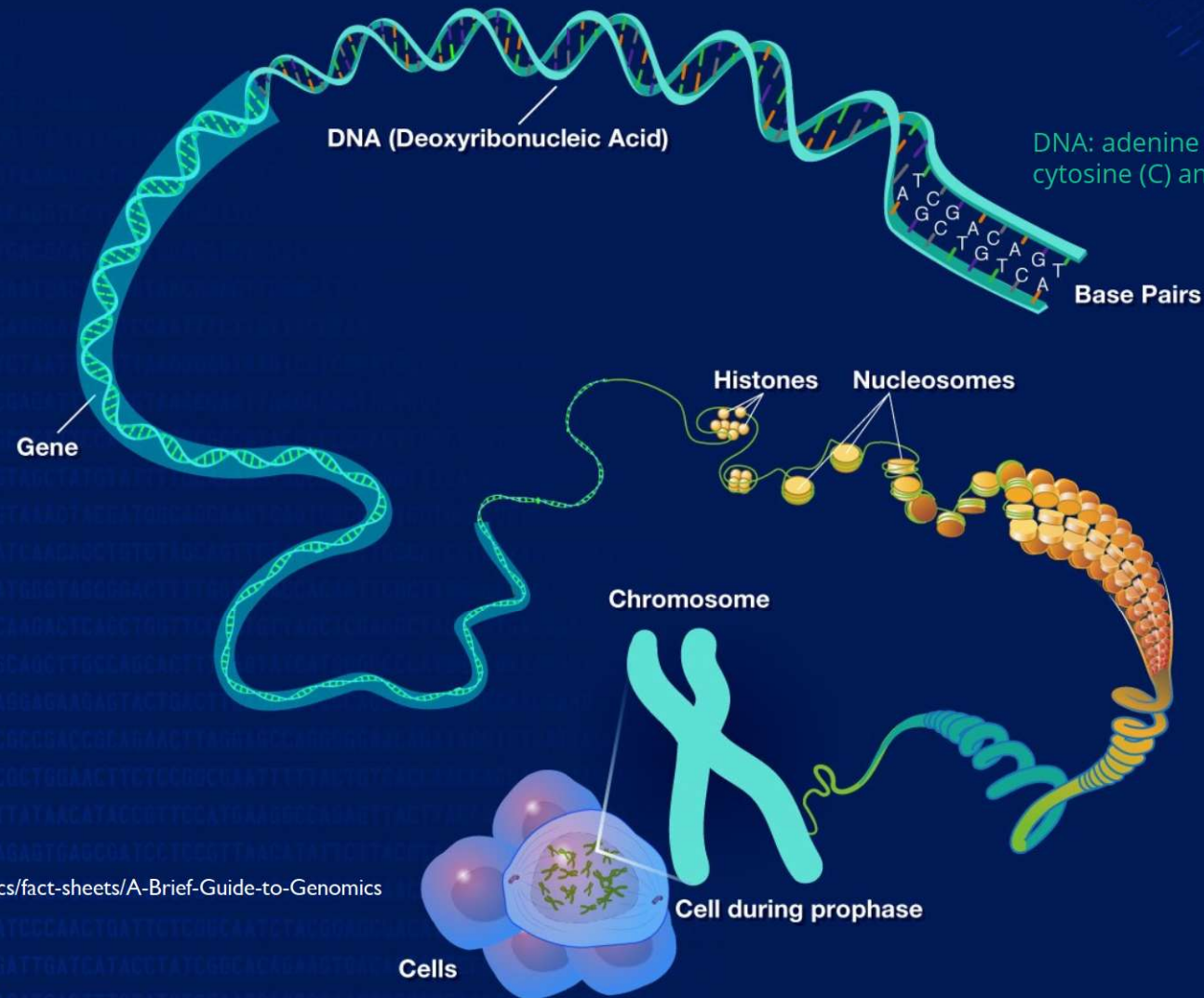


## Human genome and diseases

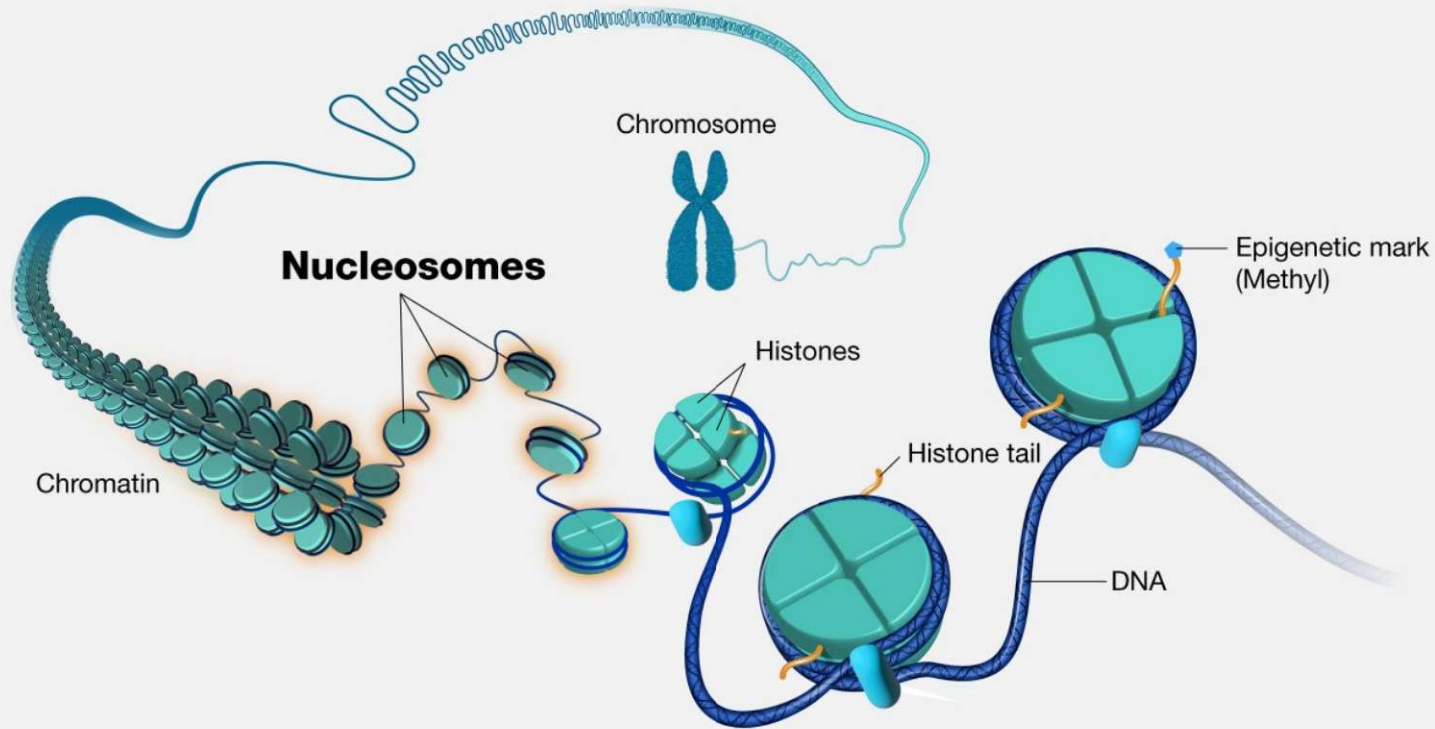


# A Brief Guide to Genomics

NHGRI FACT SHEETS  
genome.gov

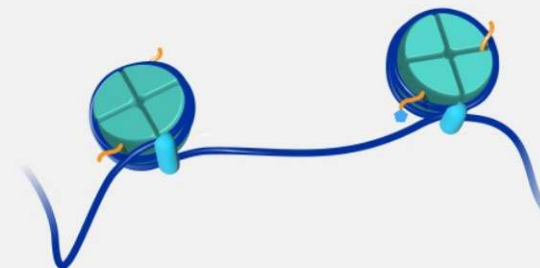
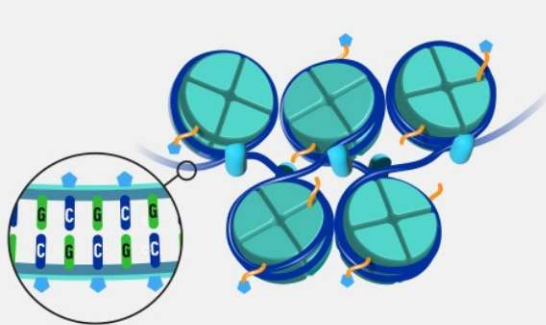


Source: <https://www.genome.gov/about-genomics/fact-sheets/A-Brief-Guide-to-Genomics>



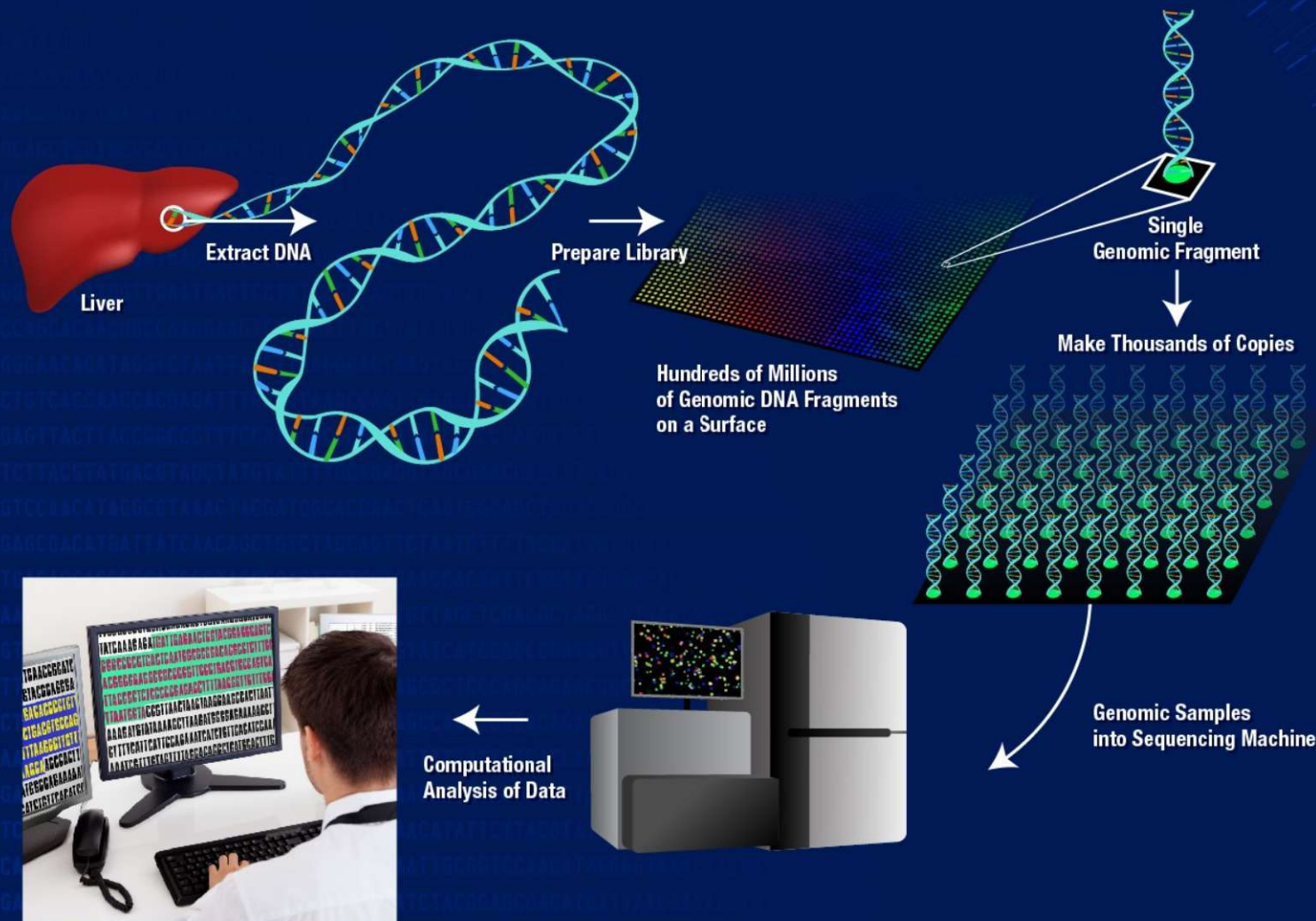
Closed chromatin (heterochromatin) is densely packed, and transcription cannot occur.

Open chromatin (euchromatin) is loosely packed, and transcription can occur.



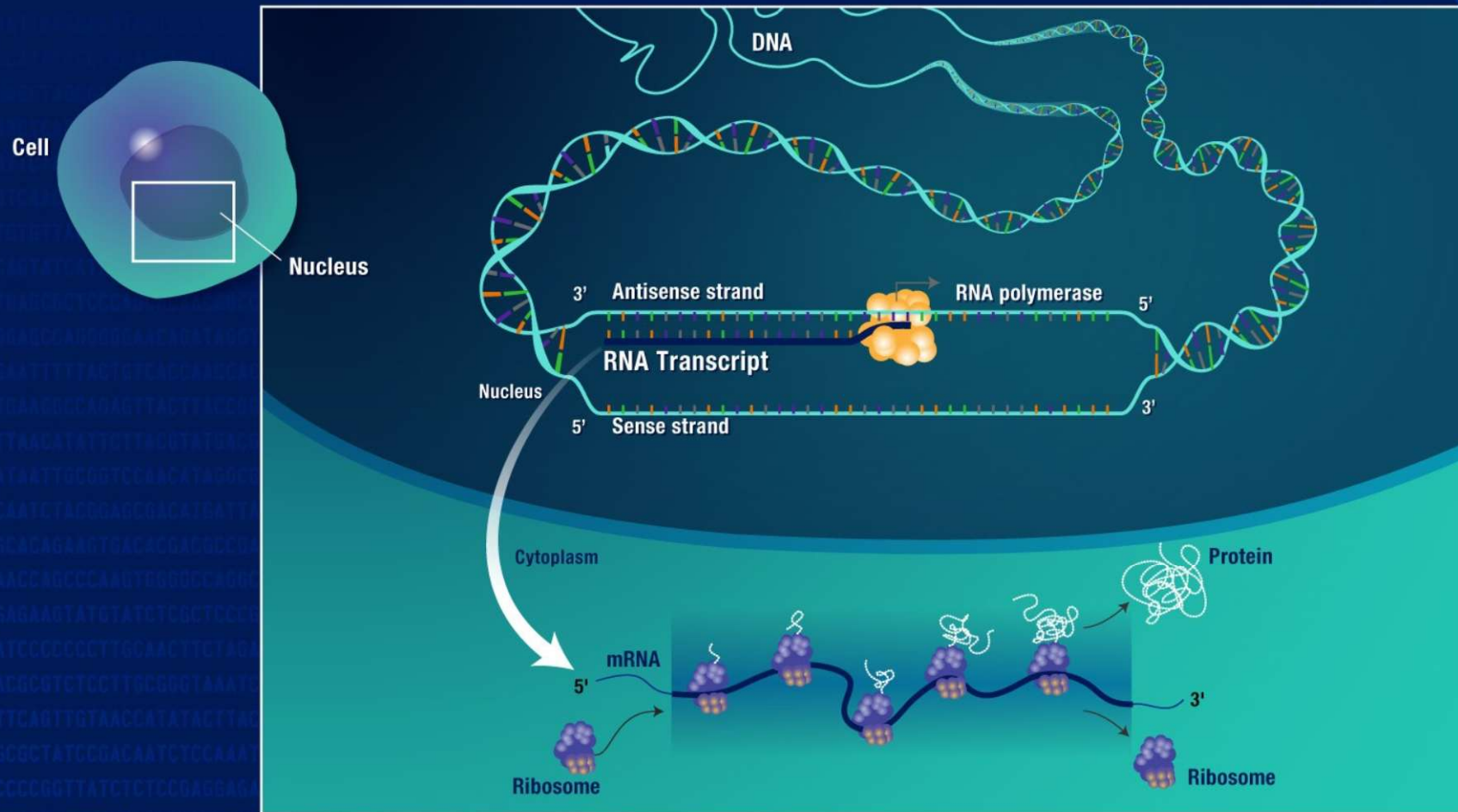
# DNA Sequencing

NHGRI FACT SHEETS  
genome.gov

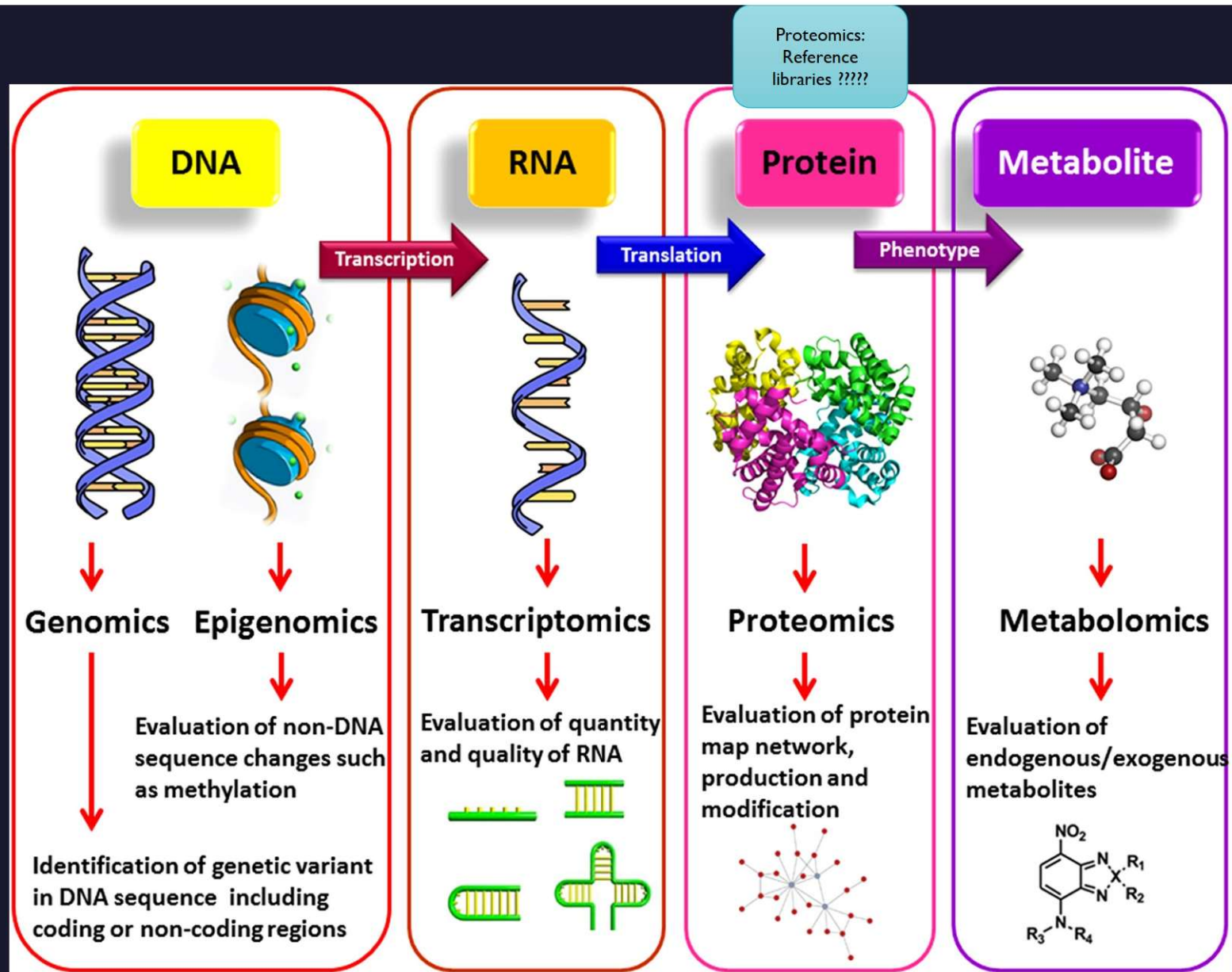


# Transcriptome

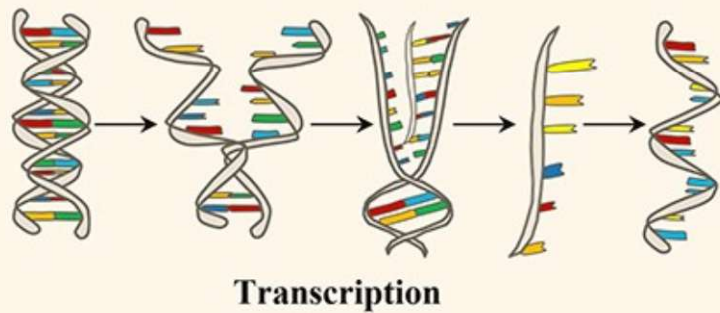
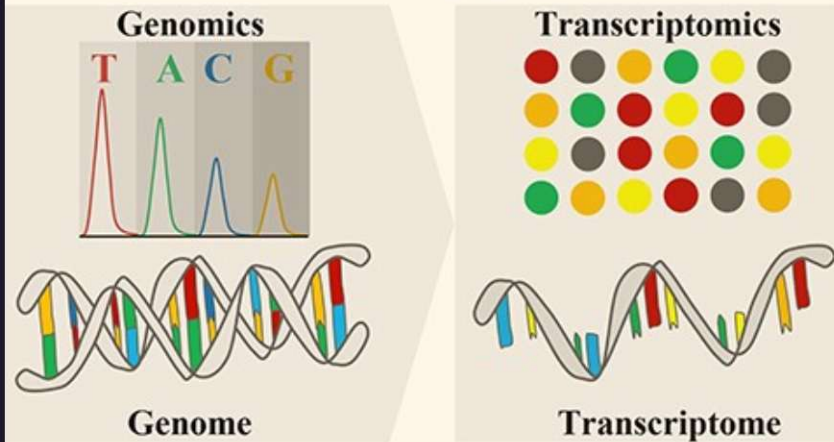
NHGRI FACT SHEETS  
genome.gov



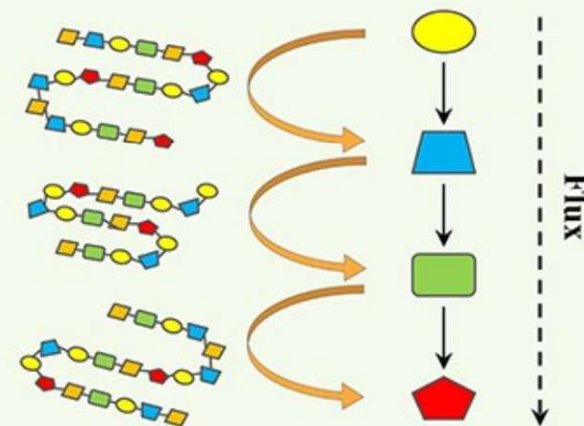
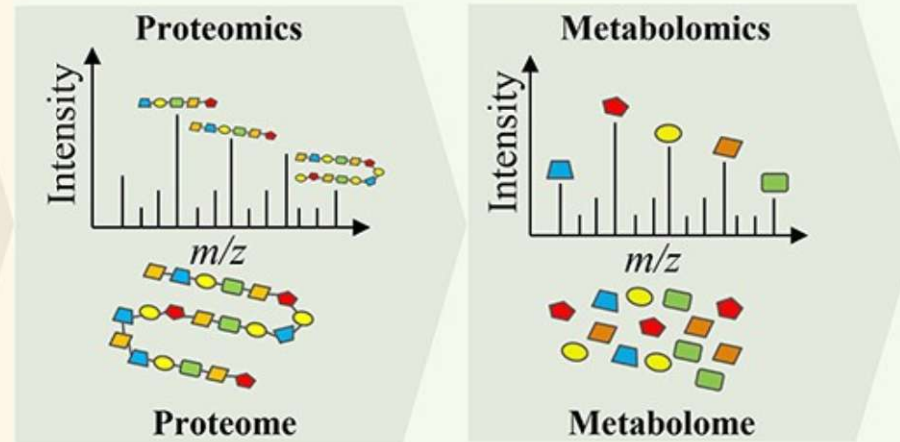
<https://www.genome.gov/about-genomics/fact-sheets/Transcriptome-Fact-Sheet>



## Genotype



## Phenotype



<https://doi.org/10.3389/fbioe.2021.613307>



# Omic

## NIH definitions



28-01-2025

@NerminGhith

**Genomics** - The genome is the complete sequence of DNA in a **cell or organism**.

**Transcriptomics** - The transcriptome is the complete set of RNA transcripts from DNA in a **cell or tissue**.

**Proteomics** - The proteome is the complete set of proteins expressed by a **cell, tissue, or organism**.

**Epigenomics** - The epigenome consists of **reversible** chemical modifications to the DNA, or to the histones that bind DNA, and produce changes in the **expression of genes** without altering their base sequence.

**Metabolomics** - The metabolome is the complete set of small molecule metabolites found within a biological sample (including metabolic intermediates in **carbohydrate, lipid, amino acid, nucleic acid**, and other **biochemical** pathways, along with **hormones** and other **signaling** molecules, as well as exogenous substances such as **drugs** and their **metabolites**).

<https://www.ncbi.nlm.nih.gov/books/NBK202165/>



# Genetic mapping

NHGRI FACT SHEETS  
genome.gov

Genetic map



Cytogenetic map



Physical map

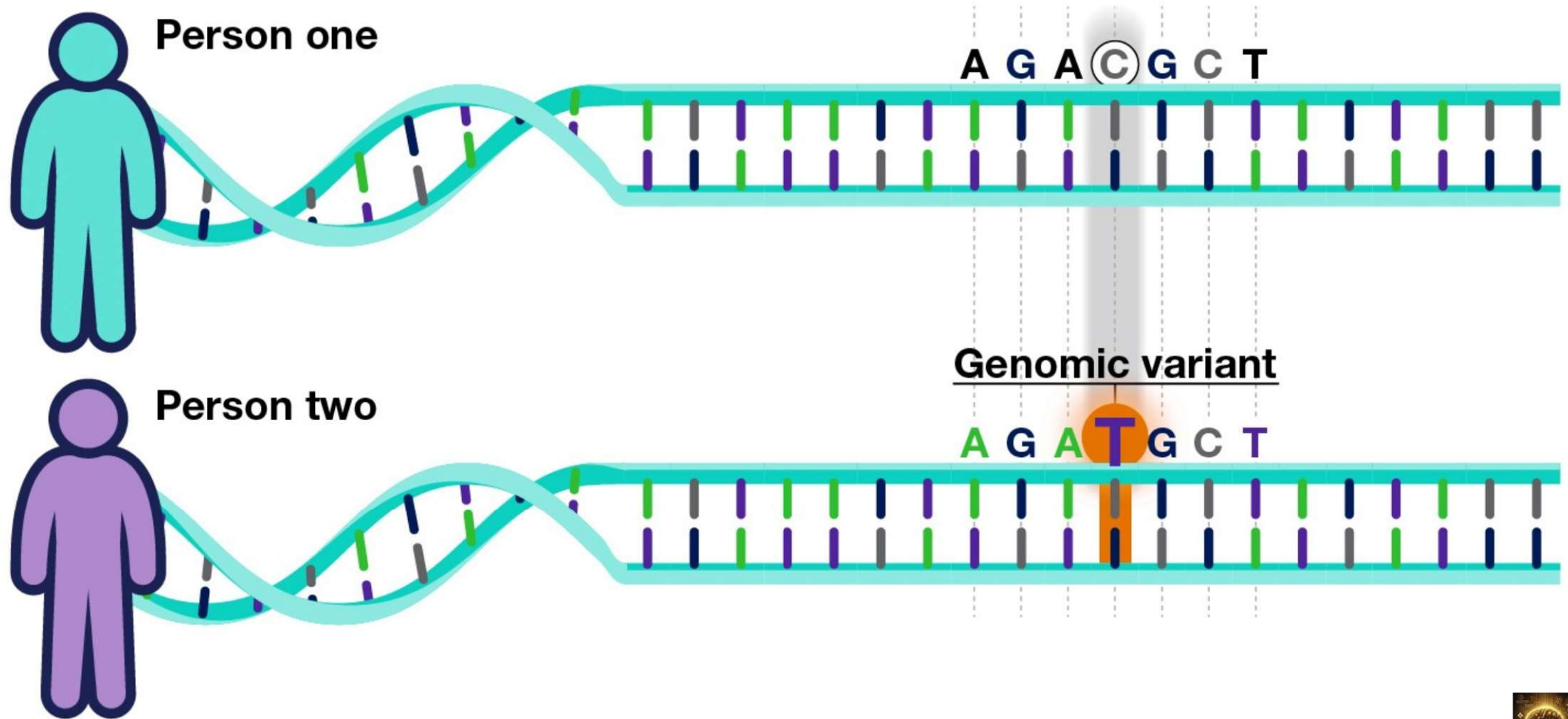


DNA sequence

....GATCTGCATGCATGCTAGCTAGCTAGCTAGAGCTTCG.... Bases

<https://www.genome.gov/about-genomics/fact-sheets/Genetic-Mapping-Fact-Sheet>

## genomic variants and the risk for specific diseases



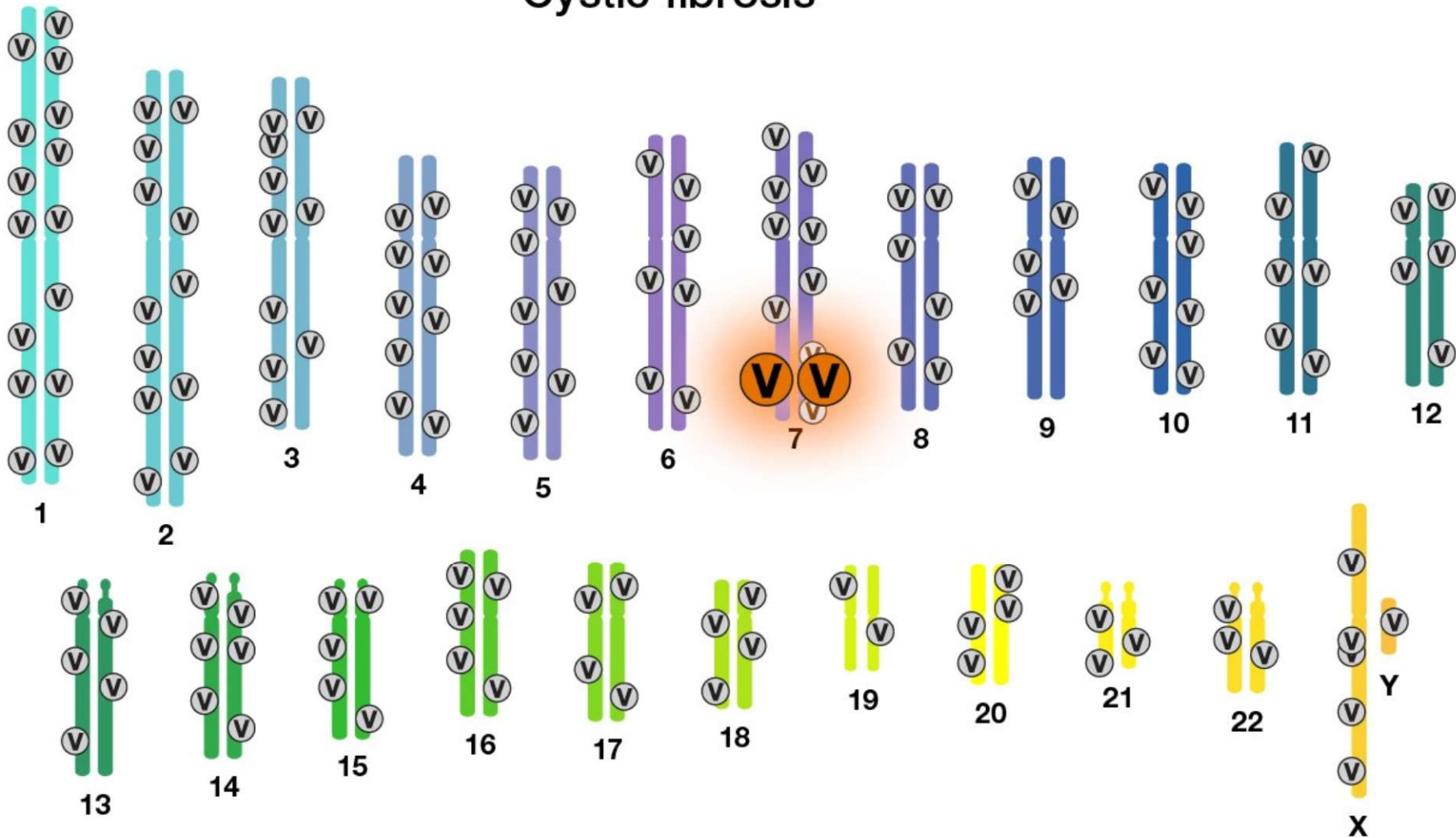
<https://www.genome.gov/Health/Genomics-and-Medicine/Polygenic-risk-scores>



Single-gene diseases

Cystic fibrosis

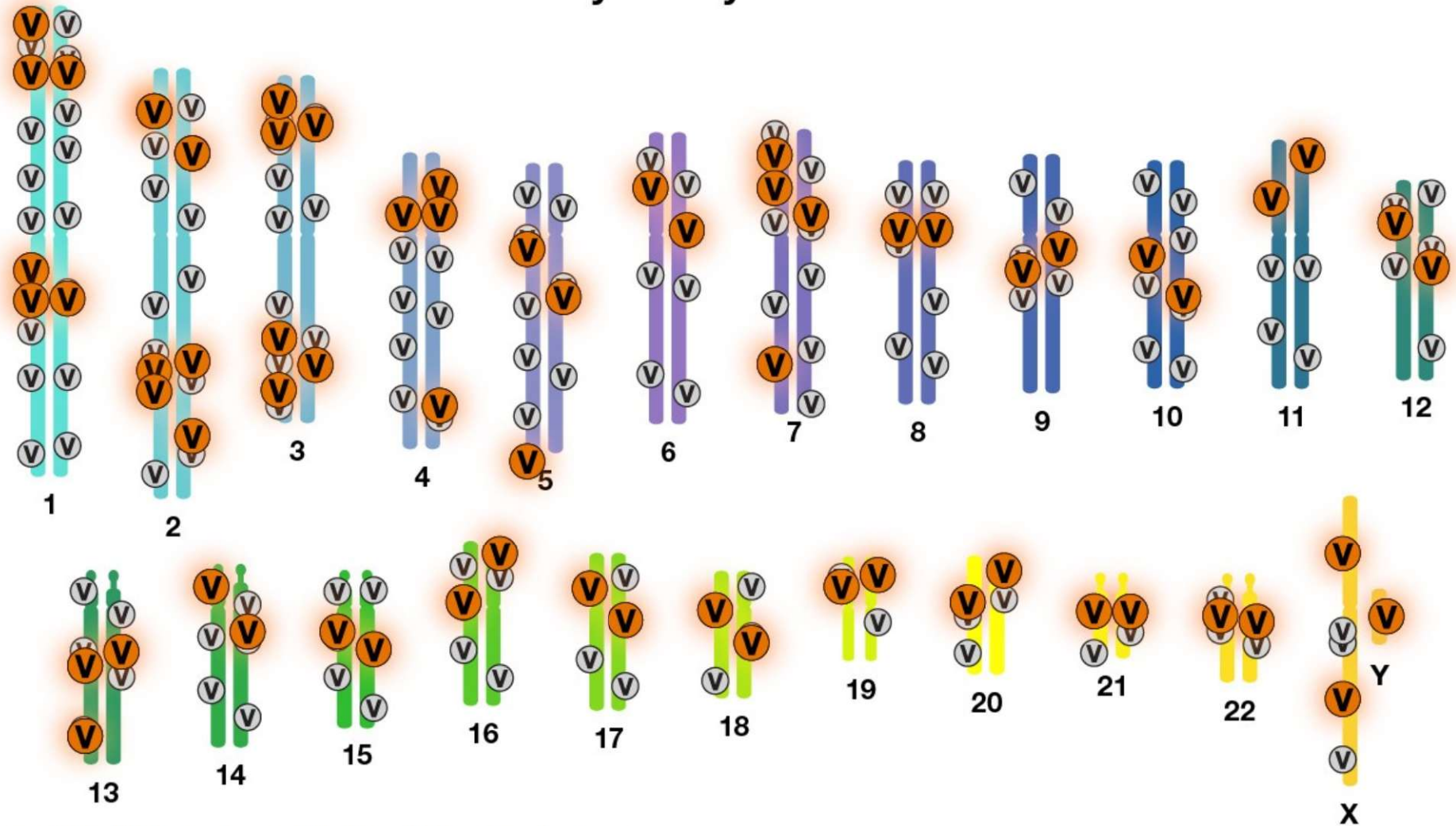
Person  
one



## Complex diseases - polygenic

### Coronary artery disease

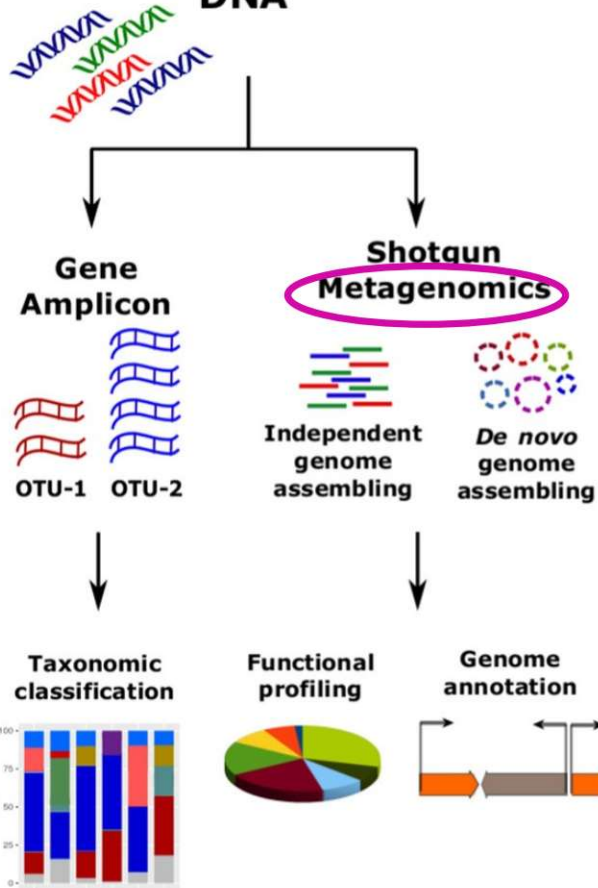
Person  
two



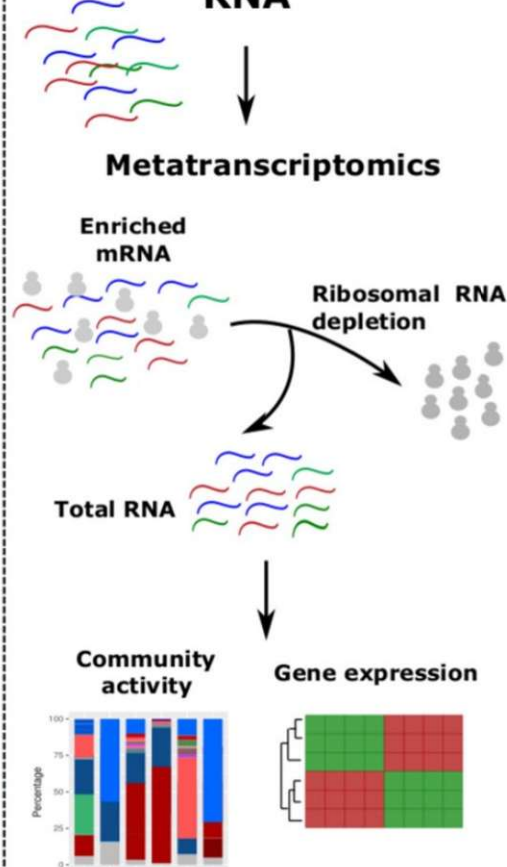
# Microbial communities

Reference libraries ?????

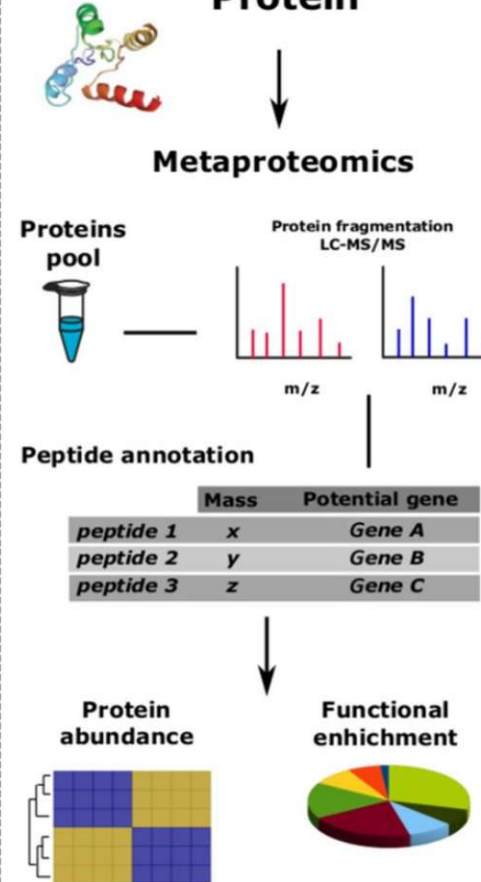
## DNA



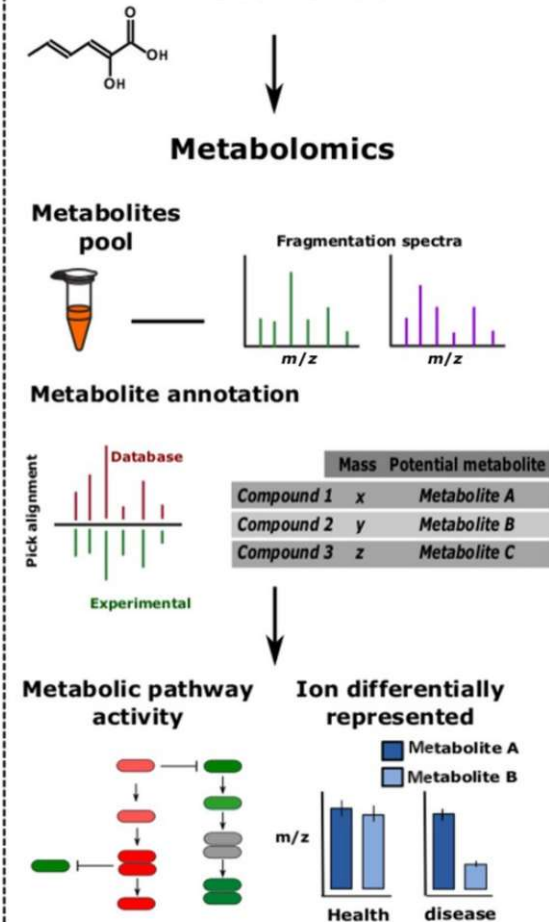
## RNA



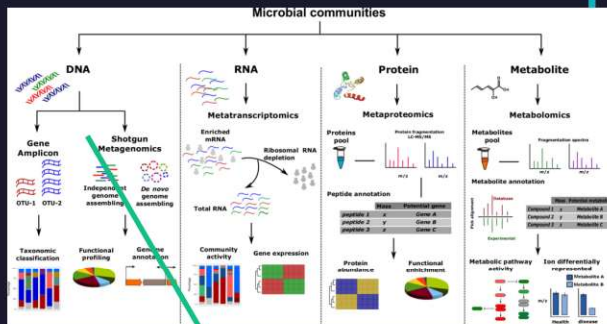
## Protein



## Metabolite



## Own Project Pipeline



Fragments of DNA

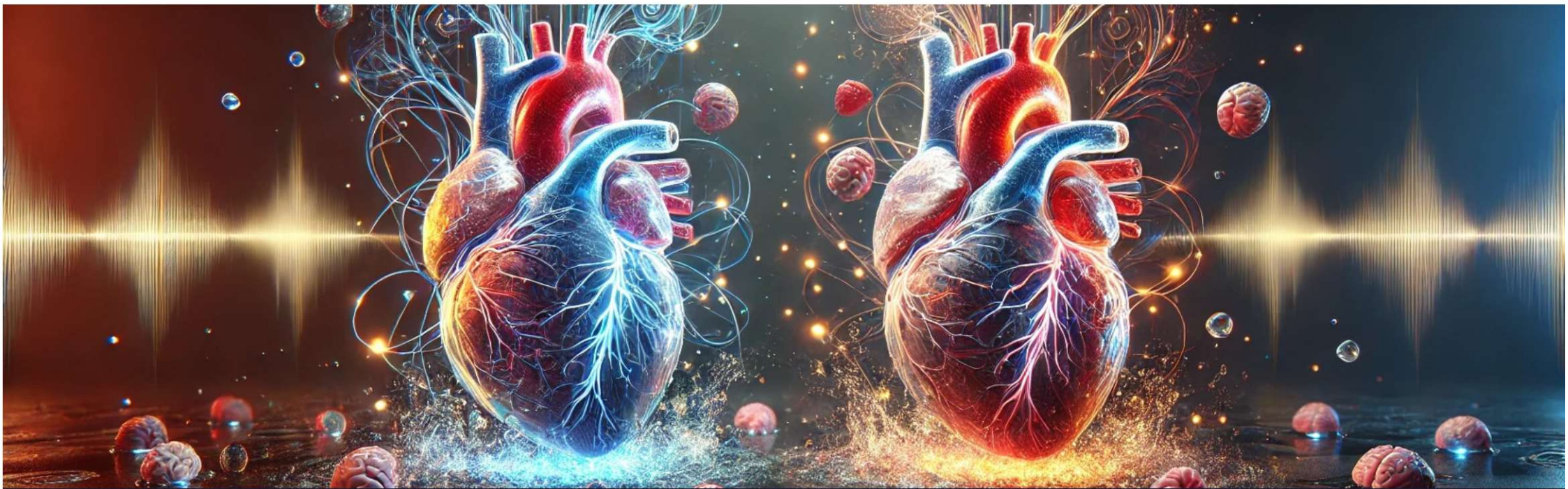
19 Classes of Antibiotics

65 Phenotypes of Antibiotics

phenotypic resistance to different classes of antibiotics

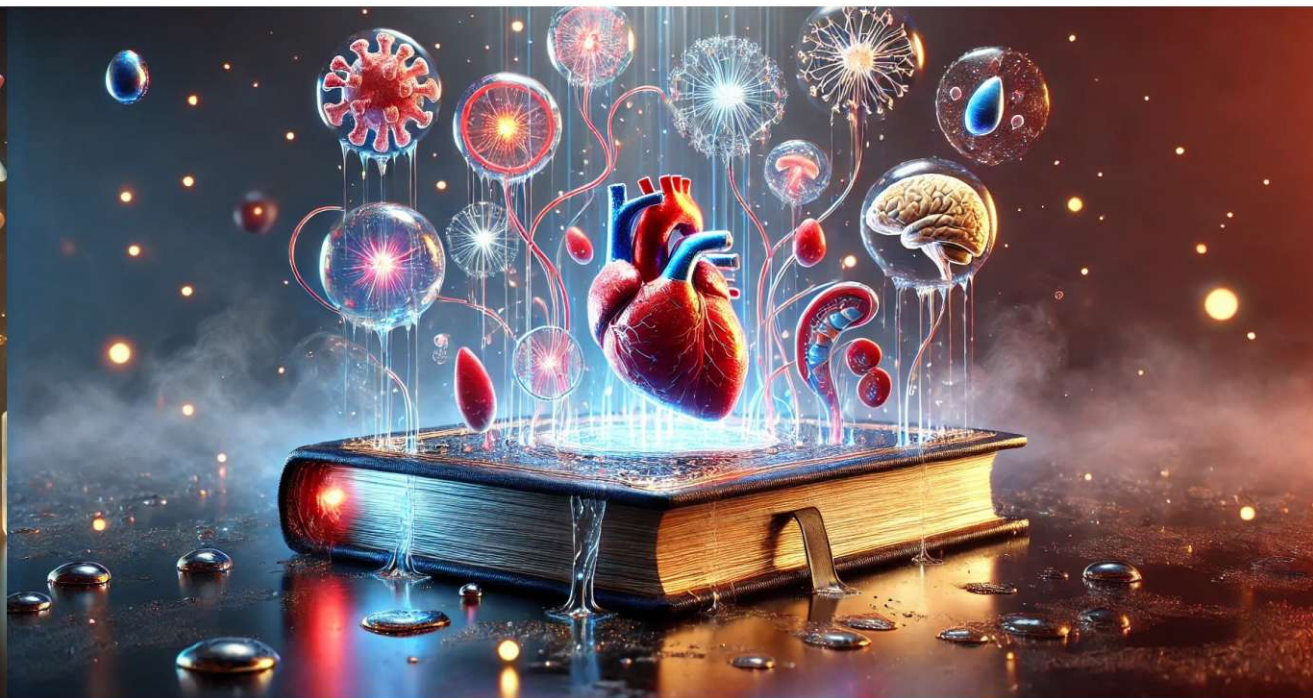
Quantify per sample (i.e., FPKM)

Resistance Genes



# Omics-based studies

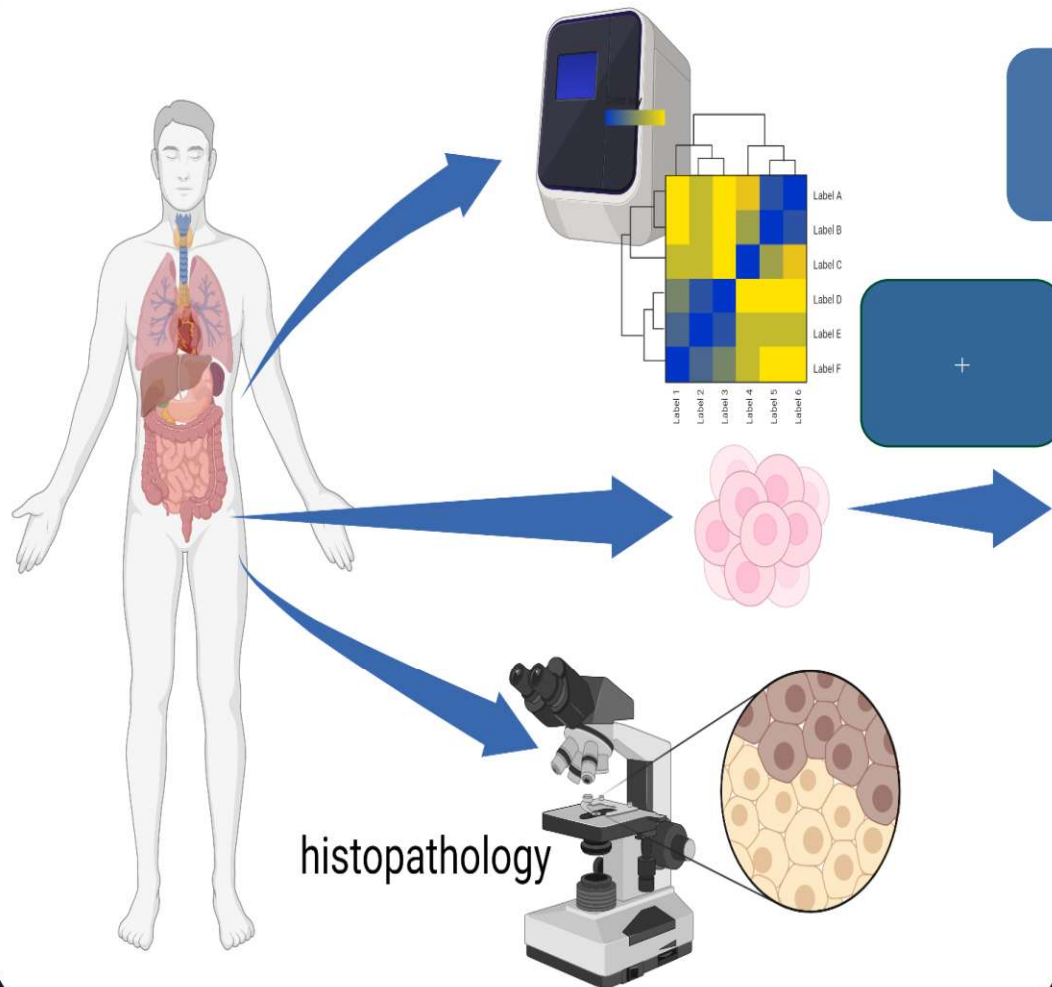
- RCTs?
- Randomization with nested cohorts or combined with an observational omics-based study?
- How can we integrate omics into a traditional RCT?



# Integrated (multimodal) Omics

- Data – modeling – classification of health events
  - Health data – models – decisions or/and predictions
- Multidata type – identification of biomarkers – clinical outcomes
- Cancer cohort –intra-tumor heterogeneity – classification of data and/or events - survival estimation – etc
- Many other frames of reference

## NGS analysis



National cancer  
registries



Central population  
registries

Matched by sex, age and  
country

> 33,160  
one-year  
survivors

> 212,892  
comparisons

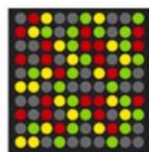
National hospital  
registries  
(discharge diagnoses  
according to ICD to  
evaluate late effects)

## Multi-data type

Genomics



Transcriptomics



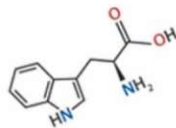
Proteomics



Epigenomics



Metabolomics



Clinical Metadata



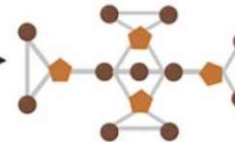
e.g., Registry data, EMR

## Biomarker Discovery

Differential Analysis



Biological Networks (GCNN, RNN, MLs)



Down-regulated subnetwork

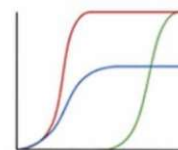
Up-regulated subnetwork

Key driver genes

Outcome



Deep Learning



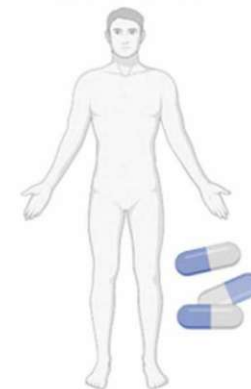
Drug A



Drug X

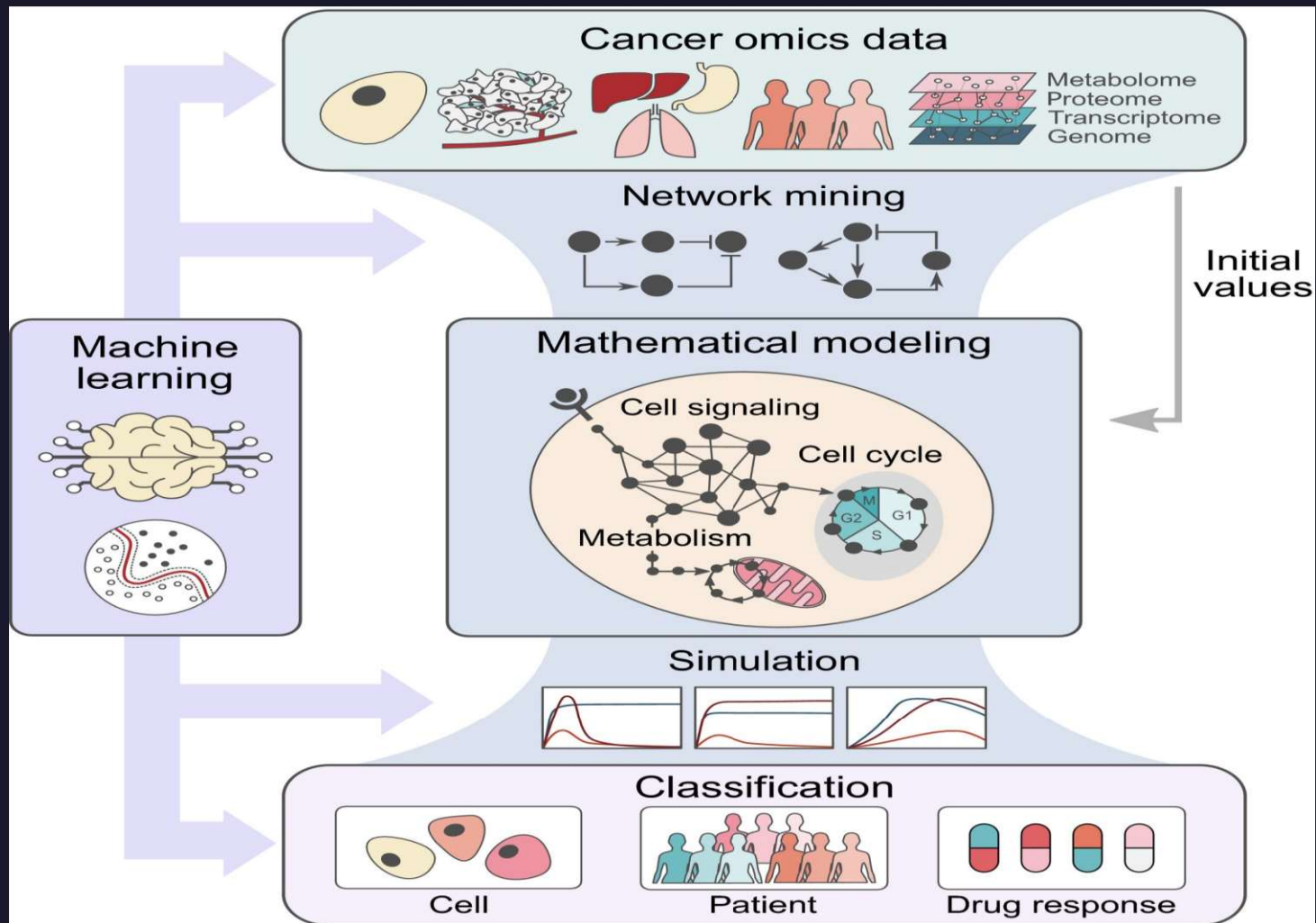
## Clinical Outcome

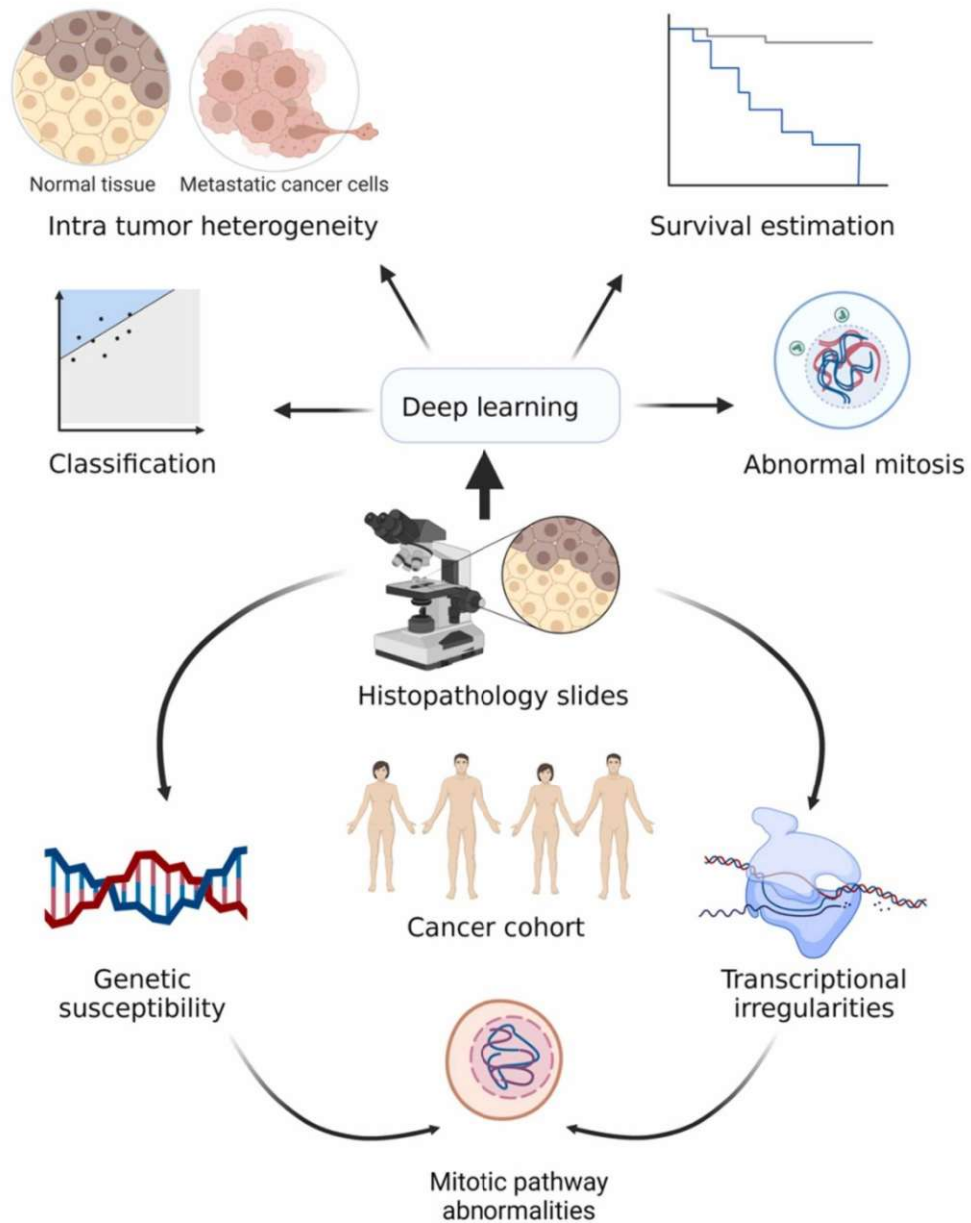
Personalized Medicine



Survival Evaluation



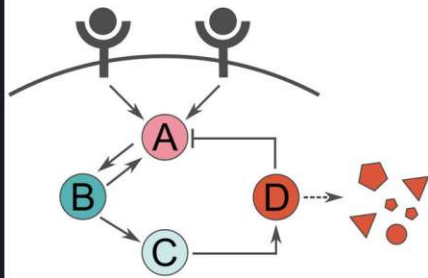




## ODE MODEL

$$\frac{dx}{dt} = S \cdot v(x, \theta), x(0) = x_0$$

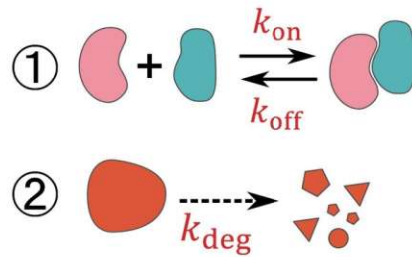
### Biological network



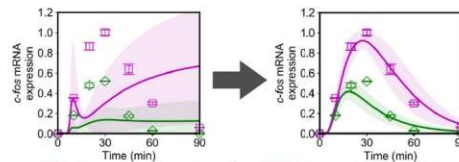
Build your own model or  
download from BioModels



### Kinetic parameters

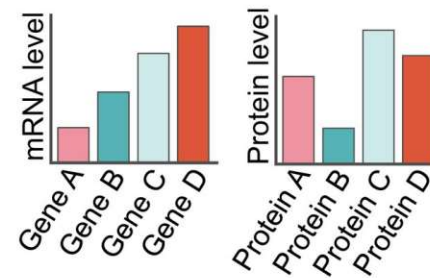


Estimate from data fitting



Fit the parameters in ODE model to data

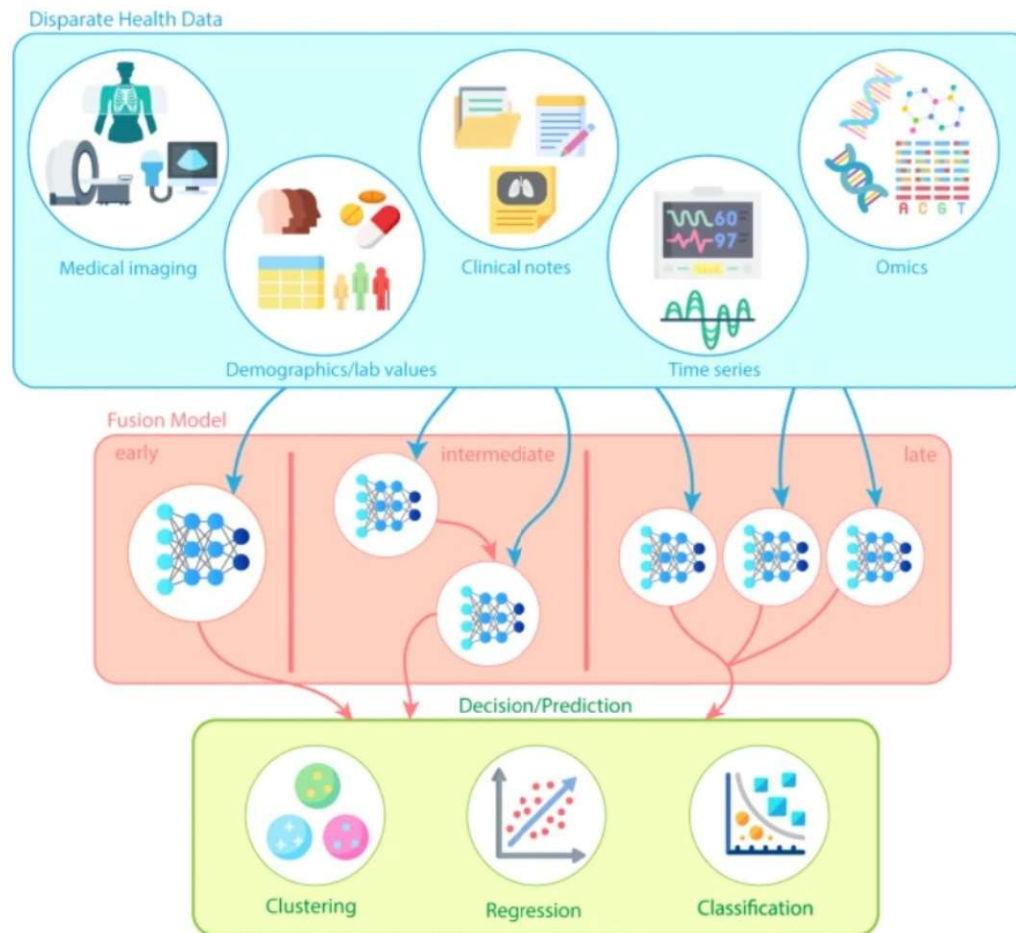
### Initial conditions



Obtain from transcriptome  
or proteome data

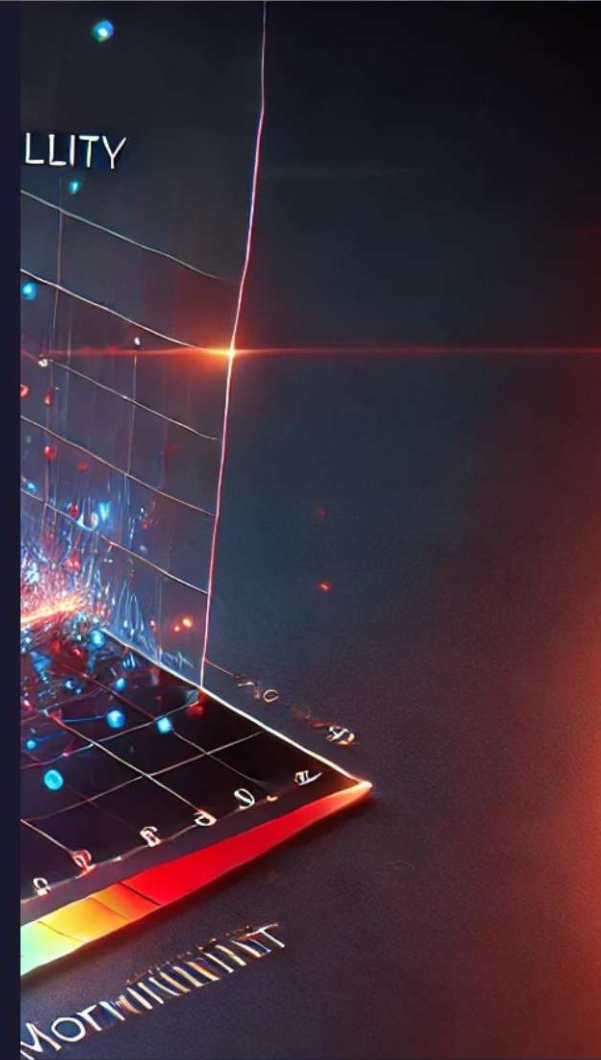


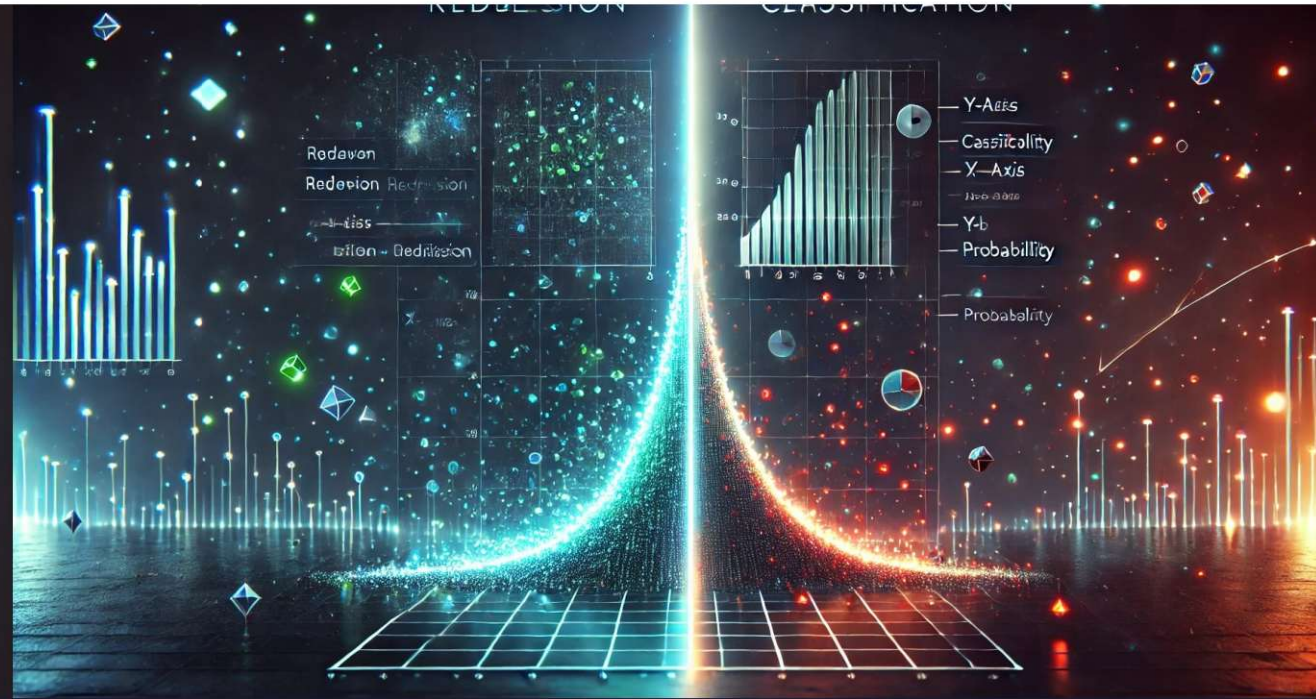
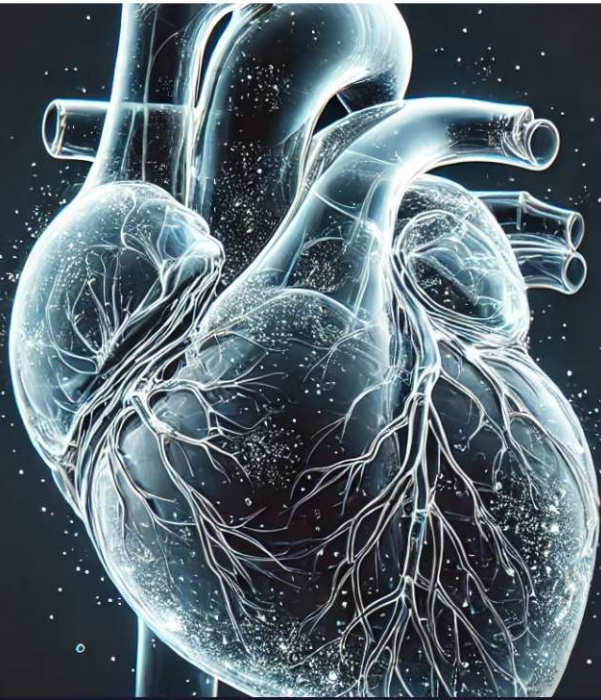
**Fig. 2: Early, intermediate, and late fusion; flow of information from information commons to model structure to outcomes.**



Information fusion can occur in a myriad of ways. In machine learning, early, intermediate, and late fusion is typified by if all the information flows into a single model (early), a step-wise fashion where outputs from one model become inputs for another (intermediate), and lastly, where all unique data types undergo separate modelling after which ensembling and/or voting occurs (late).

# Evidence pyramid and analytics





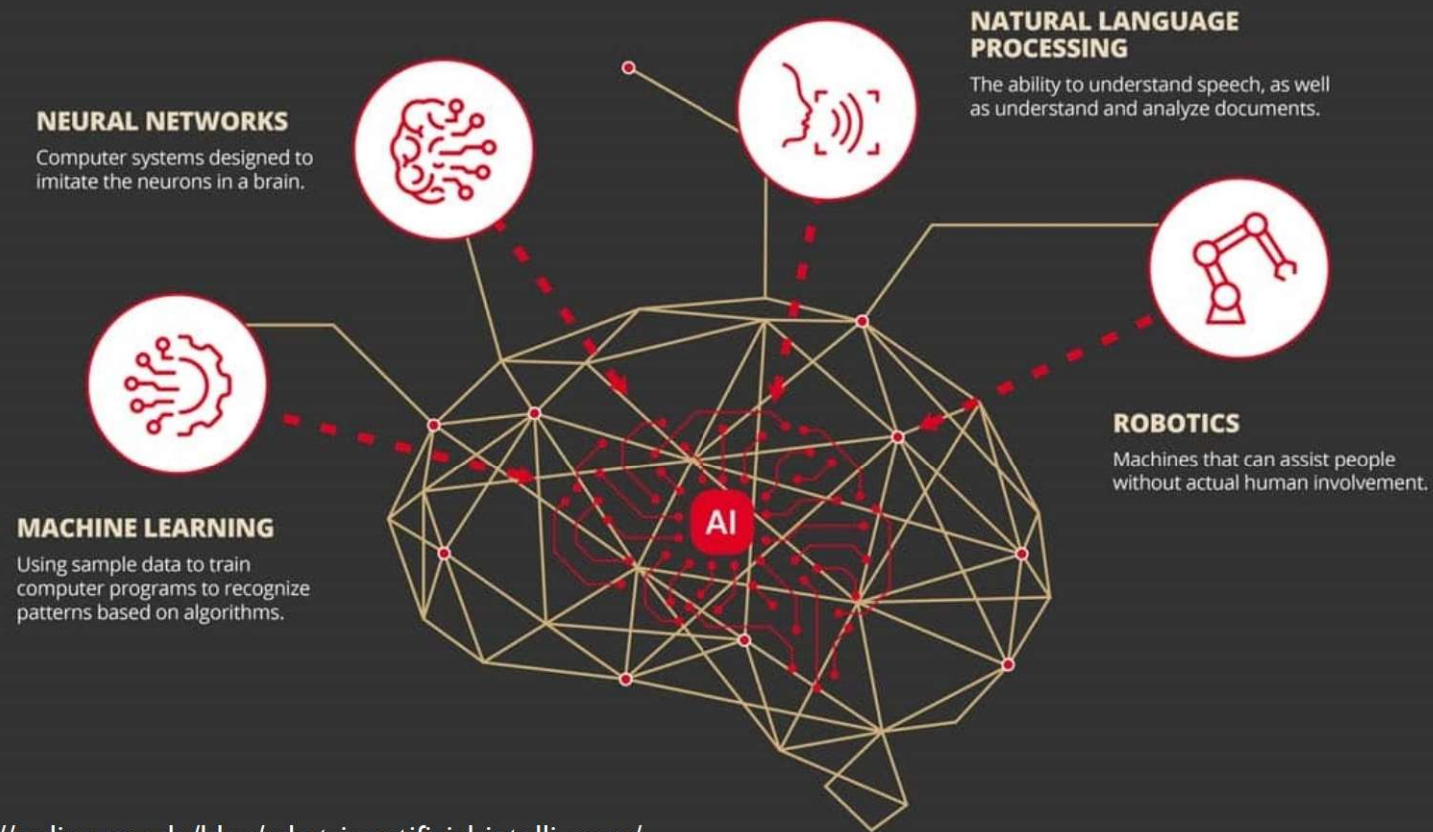
# Artificial intelligence (AI)

AI is a science and a set of **computational** technologies that are inspired by — but typically operate quite differently from — the ways people use their **nervous** systems and bodies to **sense, learn, reason, and take action**

[genome.gov](https://www.genome.gov)

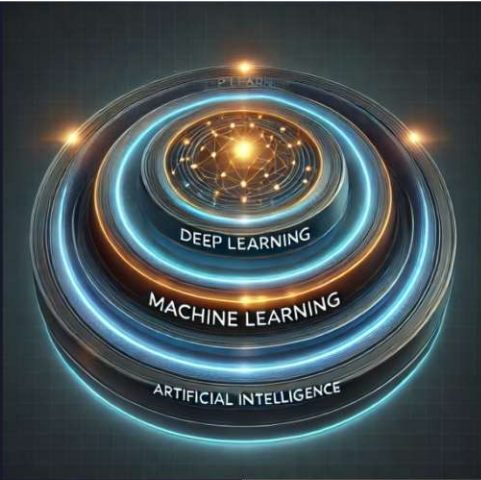


# WHAT IS ARTIFICIAL INTELLIGENCE?



<https://online.uc.edu/blog/what-is-artificial-intelligence/>





## Artificial Intelligence (AI)

AI refers to computer systems that mimic human intelligence, enabling them to solve problems and understand language.

## Machine Learning (ML)

ML is teaching systems to learn from data, enhancing performance without explicit programming.

## Deep Learning

Deep Learning employs layered neural networks to find intricate patterns, excelling in tasks like image and speech recognition.



1950's



## Artificial intelligence (AI)

*Human intelligence exhibited by machines*

1980's



## Machine learning

*AI systems that learn from historical data*

2010's



## Deep learning

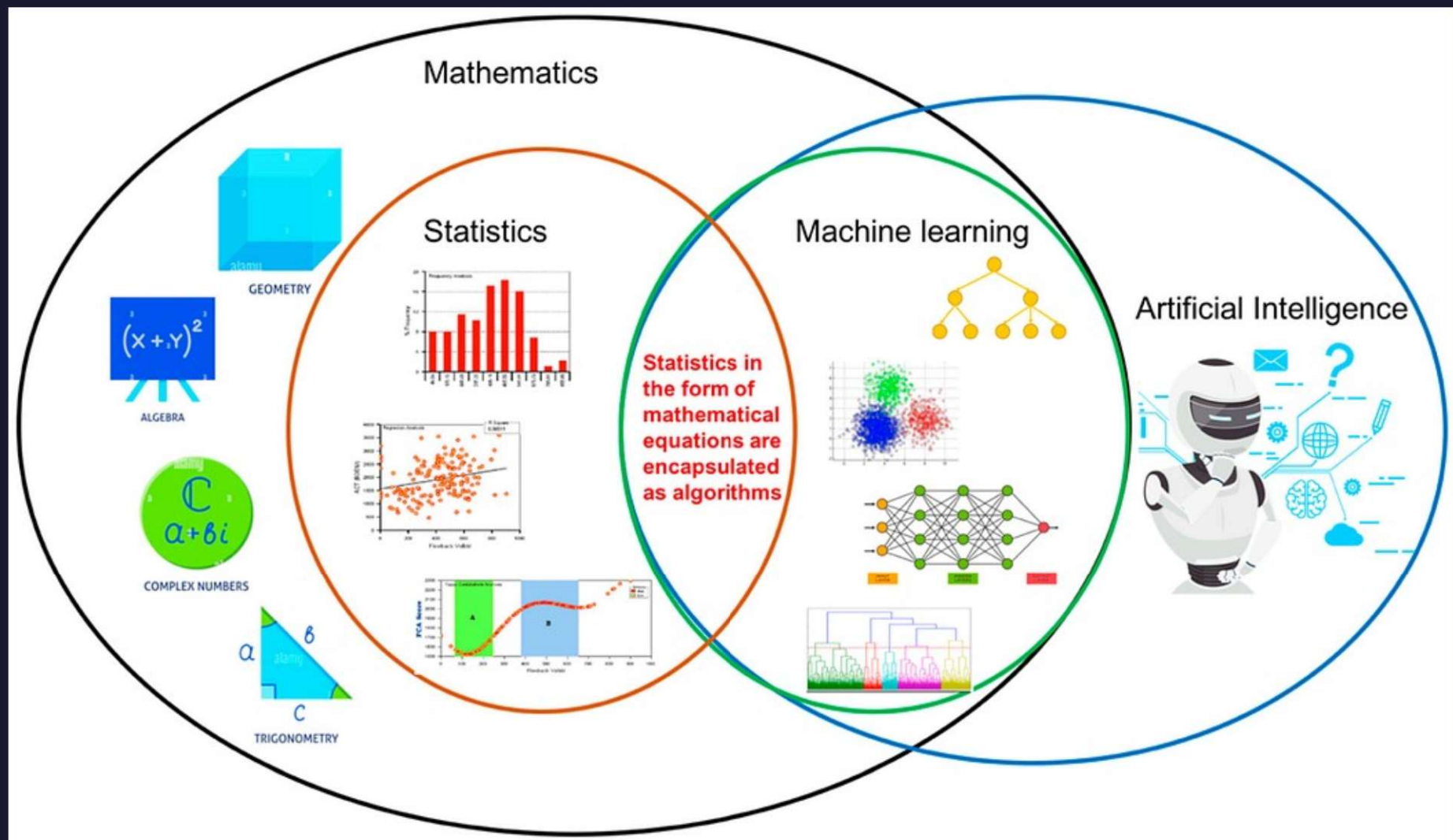
*Machine learning models that mimic human brain function*

2020's

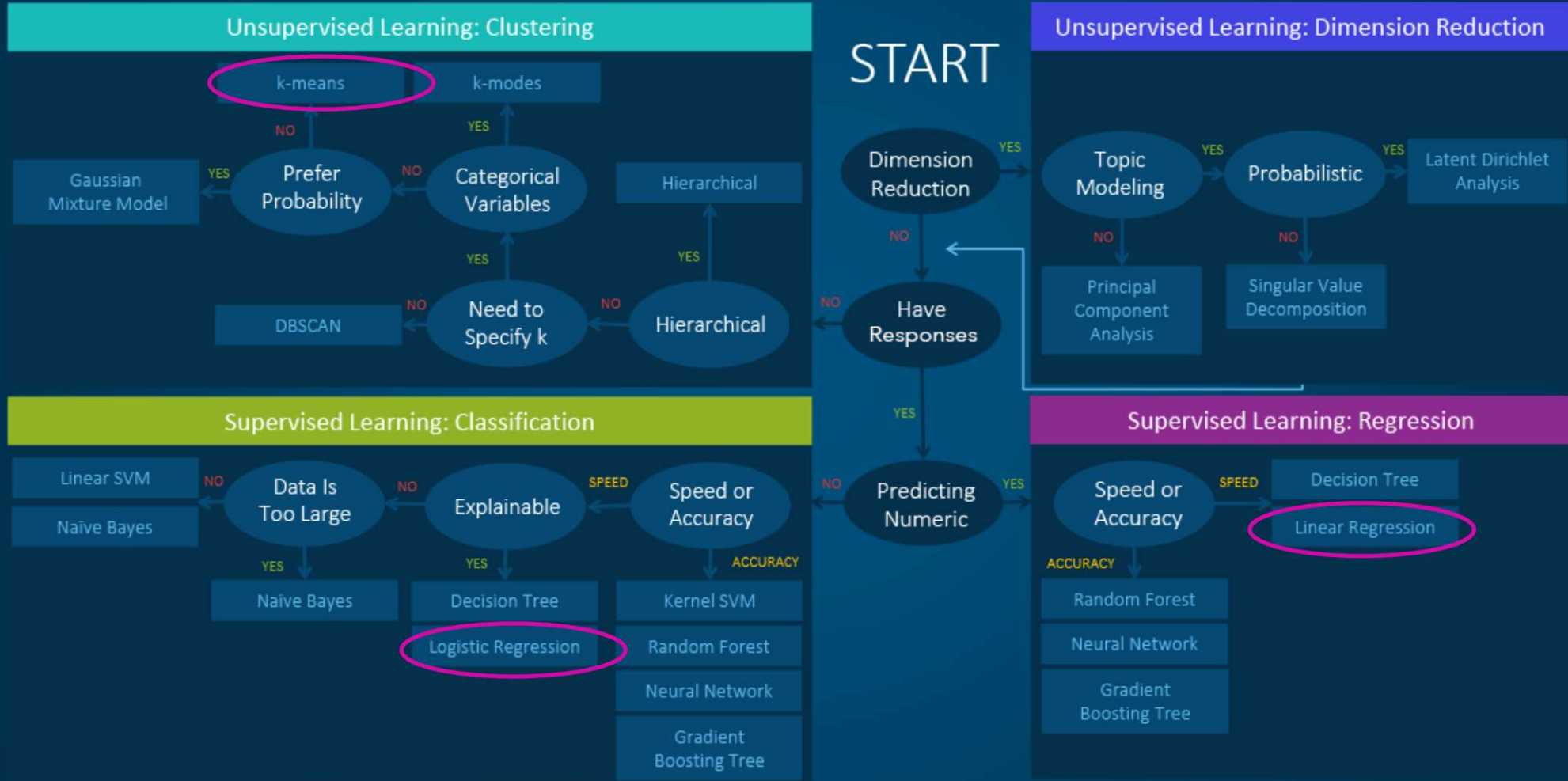


## Generative AI (Gen AI)

*Deep learning models (foundation models) that create original content*



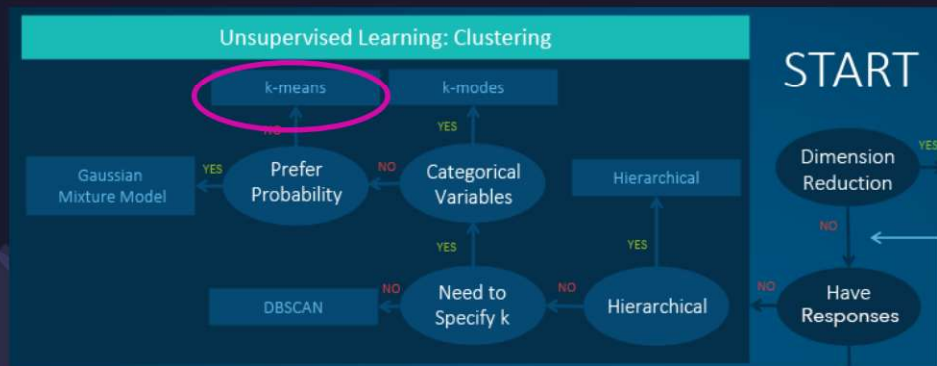
# Machine Learning Algorithms Cheat Sheet





# 2018-2019: Researcher, Research Unit of Chronic Conditions, CKFF

- Identify clusters of multimorbidity in the general population: a systematic **review** (PI-ongoing project).
  - Clusters of multimorbidity in the **Danish** population.
- Identify utilization of national **healthcare services** by individuals with multimorbidity in Denmark.



**Table 1. Chronic conditions included in definition of multimorbidity (N = 16).**

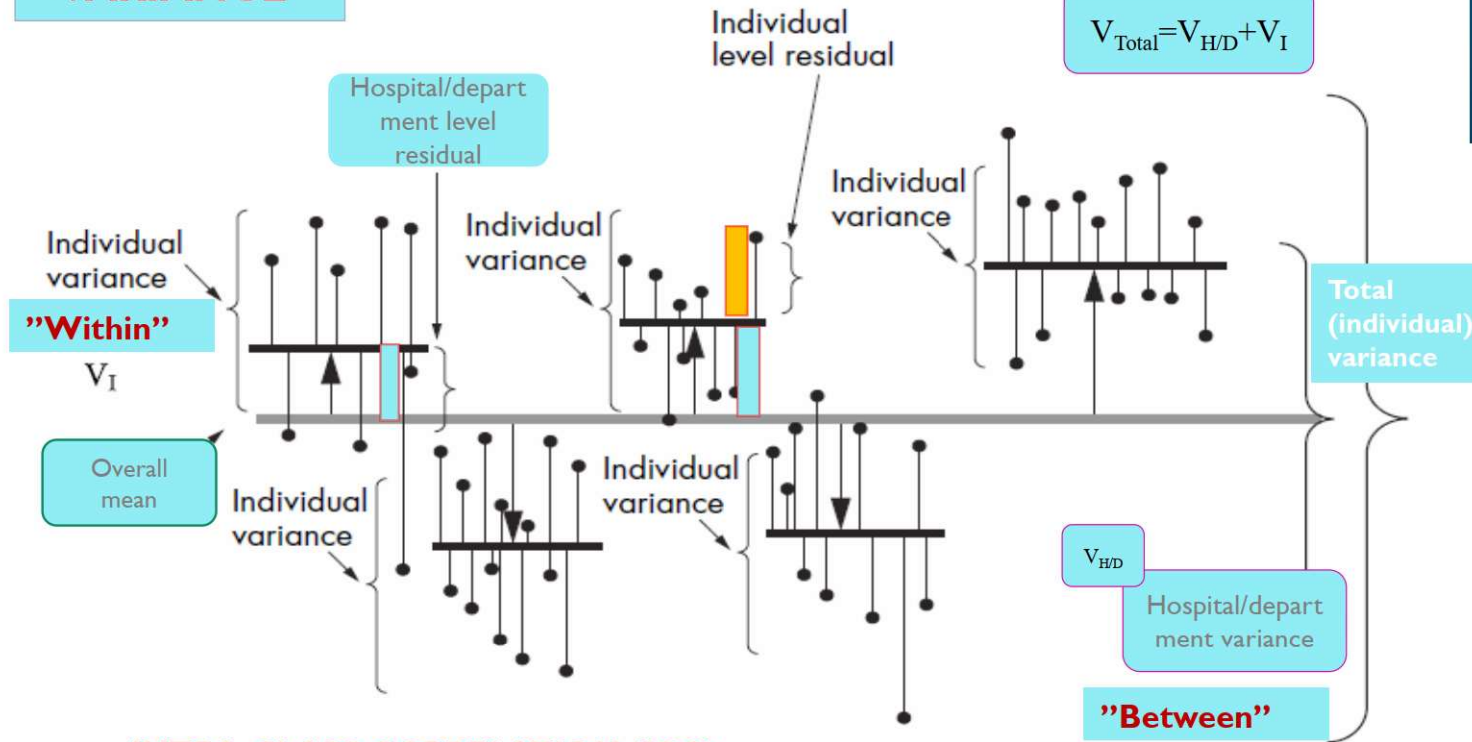
|                                       |
|---------------------------------------|
| Allergies                             |
| Hypertension                          |
| High cholesterol                      |
| Diabetes (type 1 and type 2)          |
| Heart disease                         |
| Stroke                                |
| Back pain                             |
| Joint disease                         |
| Osteoarthritis                        |
| Osteoporosis                          |
| Chronic obstructive pulmonary disease |
| Cancer                                |
| Dementia                              |
| Anxiety                               |
| Long-term use of antidepressants      |
| Schizophrenia                         |

<https://doi.org/10.1371/journal.pone.0214183.t001>

## VISUAL STATISTICS...

## Multilevel analysis

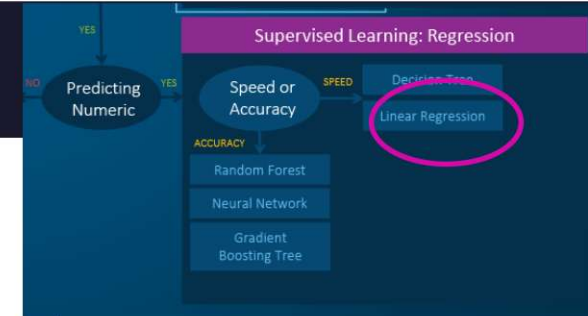
### "VARIANCE"



### INTRA-CLASS CORRELATION (ICC)

$$VPC/ICC = \frac{\text{Between } V_{\text{Hospital/department}}}{\text{between } V_{\text{Hospital/department}} + \text{within } V_I}$$

Total individual variance = within + between



Ghith and Merlo et al (2016): <https://doi.org/10.1371/journal.pone.0148187>



Machine learning is like  
statistics on steroids.



| Feature                                          | Statistics                                                                         | Machine Learning                                                                               |
|--------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Goal                                             | <b>Explain</b> relationships, hypothesis testing, and inference                    | Predict outcomes and optimize models for accuracy                                              |
| Focus                                            | Understanding <b>causality</b> , estimating parameters, deriving meaning from data | <b>Pattern</b> recognition, <b>generalization</b> , and <b>automation</b>                      |
| Causality vs. Correlation                        | <b>causal</b> inference (e.g., experimental design)                                | focuses on <b>correlations</b> – difficult to identify ‘why’                                   |
| Data Needs                                       | Works well with <b>small</b> , <b>structured</b> datasets                          | Handles massive, <b>unstructured</b> datasets (images, text, etc.)                             |
| Data Abundance                                   | Structured <b>manageable size</b> data                                             | <b>Big</b> data : video, audio, text, images, ....                                             |
| Methods                                          | <b>Regression</b> models, hypothesis testing, confidence intervals                 | Neural networks, decision trees, support vector machines                                       |
| Complexity                                       | Relatively <b>simple</b> models, <b>interpretable</b> results                      | Deep learning models with millions of parameters                                               |
| Algorithmic Advances and complex data structures | struggle with <b>high-dimensional</b> data                                         | ML techniques (e.g., random forests, deep learning)                                            |
| Feature Engineering                              | <b>Manual</b>                                                                      | <b>Automated</b> with e.g., Deep learning methods, convolutional networks in image recognition |
| Black Box Models vs. Interpretability            | <b>Interpretable</b>                                                               | Sometimes like " <b>black boxes</b> "                                                          |
| <b>Non-parametric vs. Parametric Models</b>      | Usually assume a <b>distribution</b>                                               | often uses <b>non-parametric</b>                                                               |
| Computational power                              | <b>Classical</b> computation                                                       | <b>GPUs</b> , TPUs, and distributed computing                                                  |

# From algorithm to bedside

Evidence pyramid



# Cochrane SRs Godfathers

[Archibald Leman Cochrane](#) was a Scottish physician (12 January 1909 – 18 June 1988)

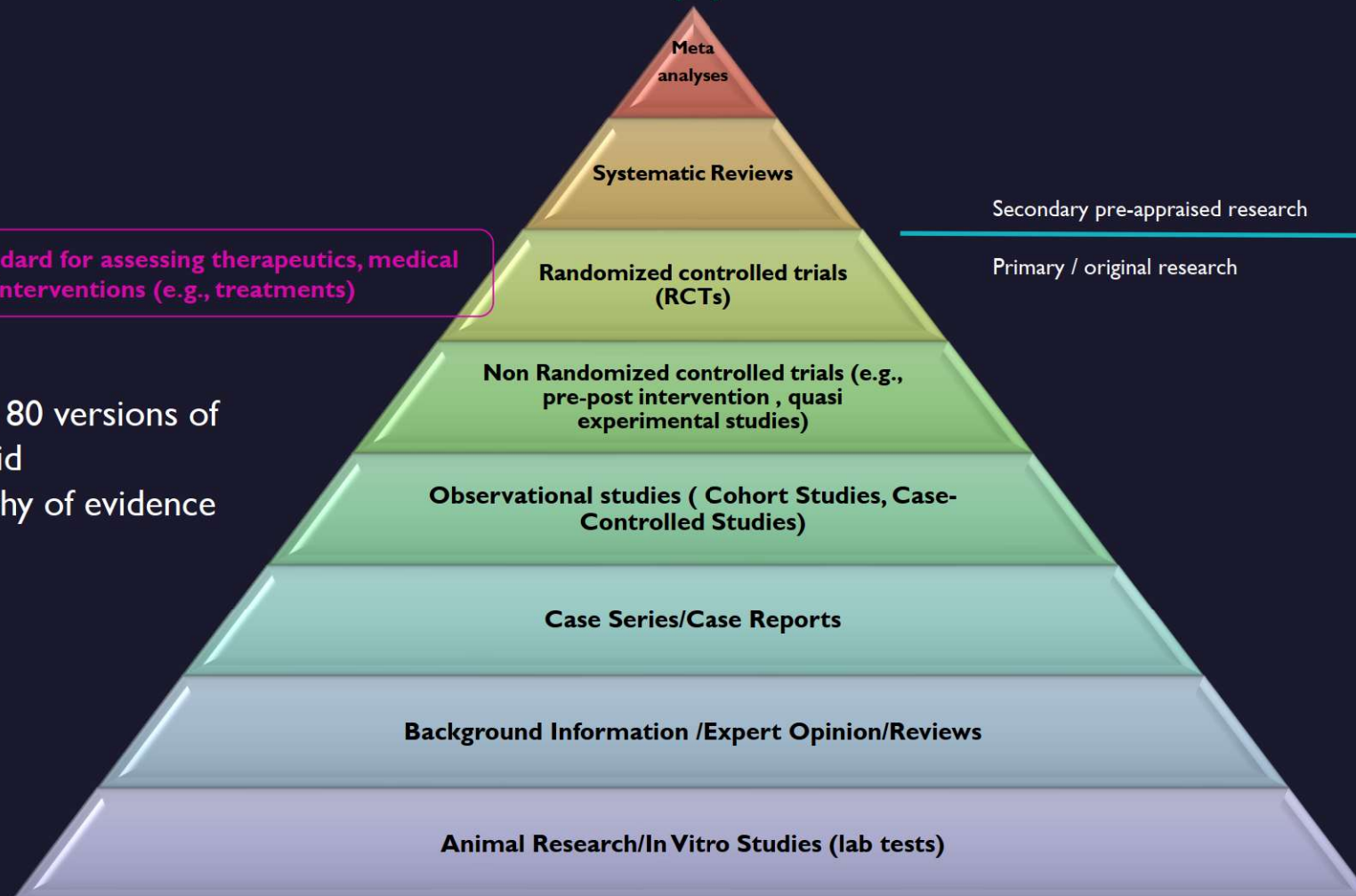


- Effectiveness and Efficiency: Random Reflections on Health Services
- Adopt [randomized](#) controlled trials to support the development of less biased [clinical trials](#) and medical interventions.

[Sir Iain Geoffrey Chalmers](#) – [Cochrane Foundation](#)



# Evidence pyramid



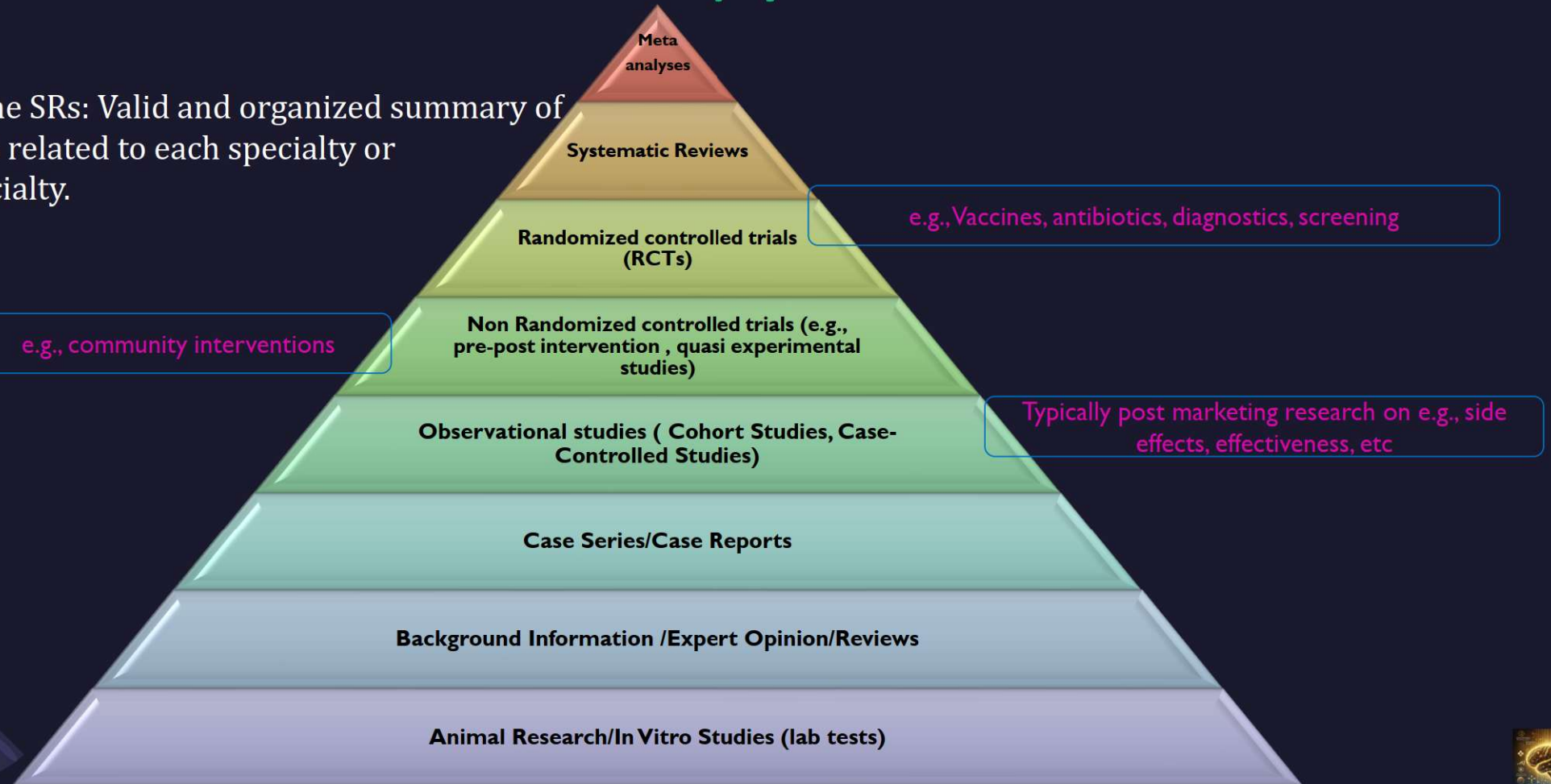
More than 80 versions of  
this pyramid  
Or hierarchy of evidence

Good reference on critical appraisal tools: <https://www.cebm.ox.ac.uk/resources/ebm-tools/critical-appraisal-tools>



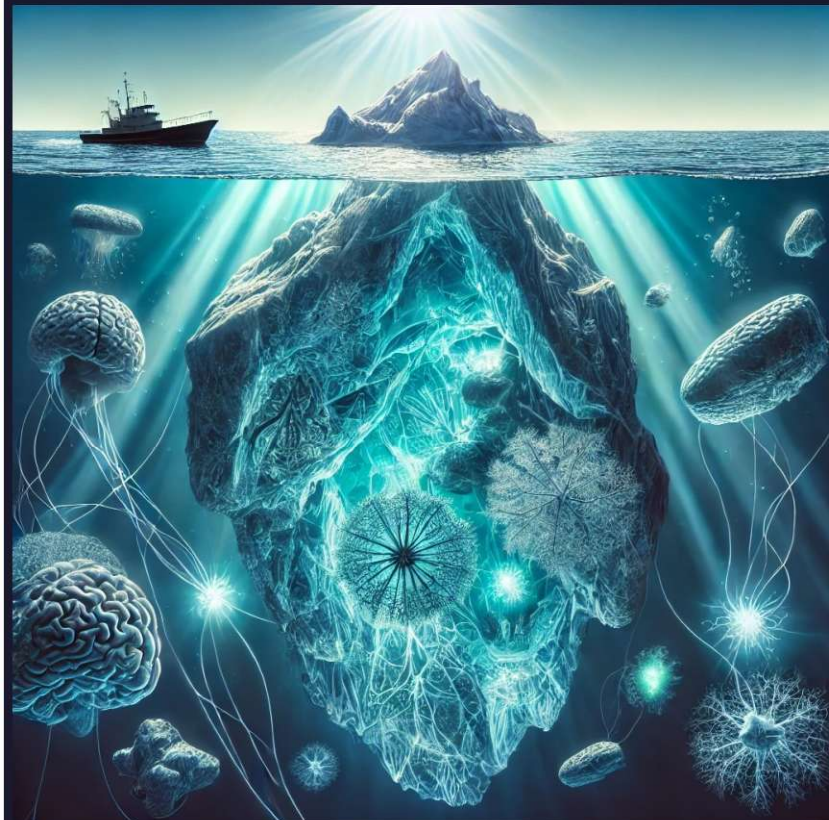
# Evidence pyramid

Cochrane SRs: Valid and organized summary of all RCTs related to each specialty or subspecialty.



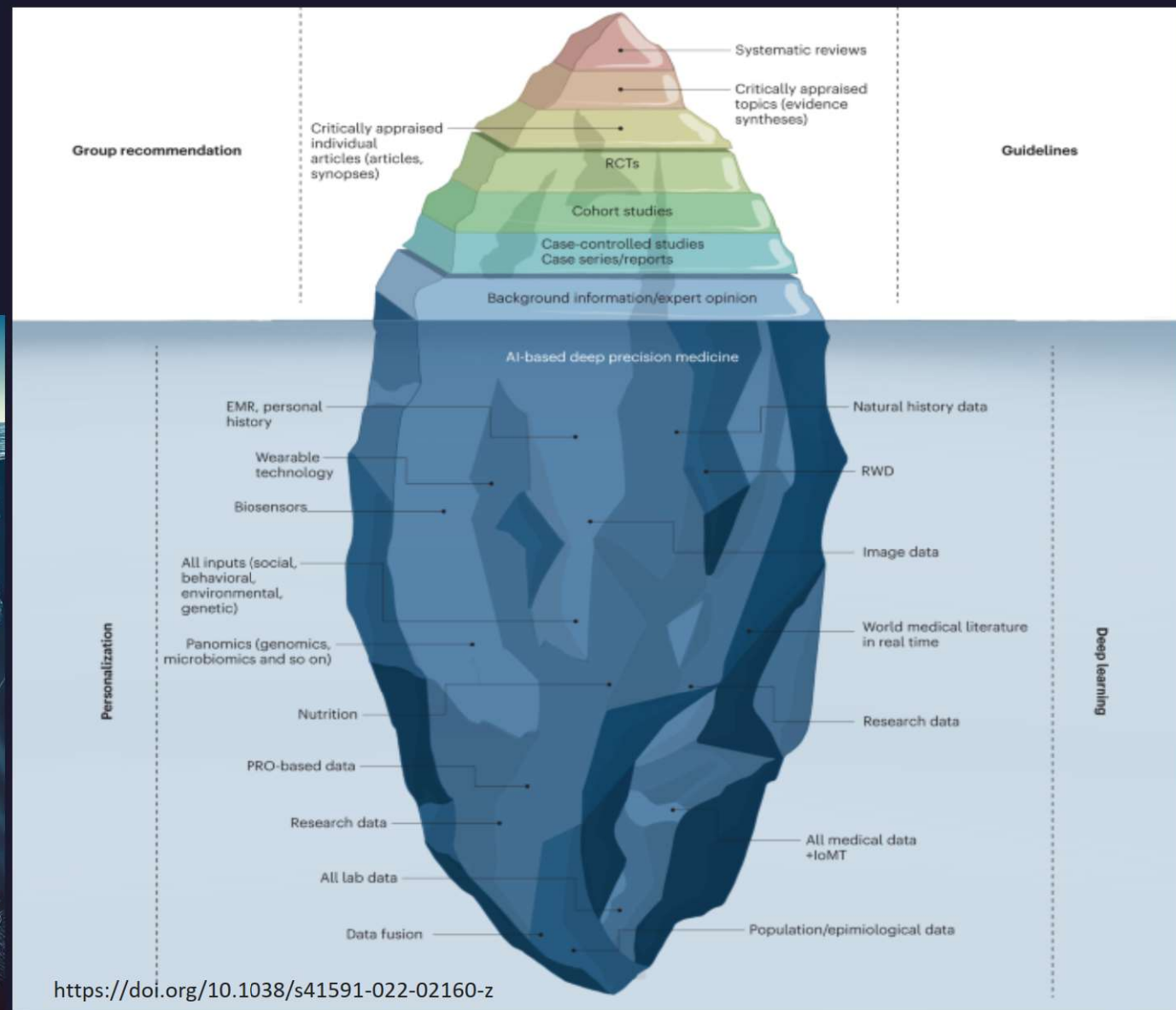


# Evidence iceberg



28-01-2025

@NerminGhith





# Thank You

Linkedin connect!

